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PROJECT REPORT
ON

"HUMAN FACIAL RECOGNITION FOR PEOPLE WITH AND WITHOUT MASK: CONTACTLESS BIOMETRICS"

Submitted in the partial fulfilment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING
IN
INFORMATION SCIENCE AND ENGINEERING

For the Academic Year 2021-2022

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2021-22

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CERTIFICATE

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ABSTRACT

While India has steadily moved towards unlocking, battling the pandemic needs a unified effort by the authorities at one level and citizens at another level to help control the spread. There is global evidence that face masks and maintaining social distance help in slow down the disease spread. To create safe environment that contributes to public safety, we propose an efficient computer vision-based approach focused on the real-time automated monitoring of people to detect both safe social distancing and face masks in public places by implementing the model using machine learning tools to monitor activity and detect violations through camera. Furthermore, many public service providers have made compulsory to use the service only if they wear masks and follow safe social distancing. Therefore, face mask detection and safe social distance monitoring has become a crucial computer vision task to help the global society. The manual monitoring method would cost more, requires much time and may not be accurate. So, the fast and better model is needed to monitor. This can be done by setting up the frame wherever we want to monitor and then detects humans in rectangular boxes and also faces in small rectangles and then system alerts if the distance is not maintained and if not wearing mask. This solution can be used in places like temples, shopping complex, metro stations, airports, etc.

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Chapter 1 INTRODUCTION

The spread of COVID-19 Pandemic Disease has created a most crucial global health crisis of the world that has had a deep impact on humanity and the way we perceive our world and our everyday lives. According to the World Health Organization (WHO as of December 30, 2020) report, the current outbreak of COVID-19, has infected over 2,13,39,853 people and more than 5,71,659 deaths in more than 200 countries around the world, carrying a mortality of approximately 2·7%, compared with a mortality rate of less than 1% from influenza. A novel coronavirus has resulted in person-to-person transmission but as far as we know, the transmission of the novel coronavirus causing coronavirus disease 2019 (COVID-19) can also be from an asymptomatic carrier with no covid symptoms. Till now there is no report about any clinically approved antiviral medicine or vaccines that are effective against COVID-19. It has spread rapidly across the world, bringing massive health, economic, environmental and social challenges to the entire human population. At the moment, WHO recommends that people should wear face masks to avoid the risk of virus transmission and also recommends that a social distance of at least 2m be maintained between individuals to prevent person-to-person spread of disease.

1.1 Introduction to Machine Learning

Machine learning (ML) is the study of computer algorithms that improve automatically through experience. It is seen as a subset of artificial intelligence. Machine learning algorithms build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so. Machine learning algorithms are used in a wide variety of applications, such as email filtering and computer vision, where it is difficult or unfeasible to develop conventional algorithms to perform the needed tasks. In our project we are training the system by giving images of different categories like without mask and with mask and then asking it to detect face mask. And also computes the reliable statistics of people maintaining social distance and wearing face mask by learning day by day and predicts the possible count on next day. It is nothing but making machines learn and think and act like humans. It is the process of enabling machines to learn from past experiences and to improve the accuracy of outputs over time. It is hard for anyone to program each and every task, right? But with Machine Learning, you do not need to program each task; instead, the computer itself develops an algorithm according to the task to be performed using the available data.

1.2 Applications of Machine Learning

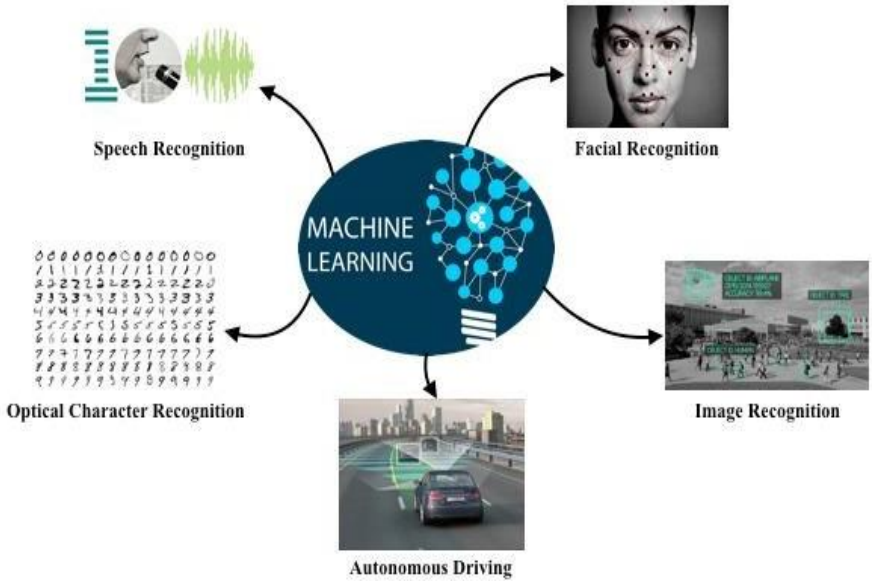


Fig. 1.1 Applications of Machine Learning

Here are the few applications of ML,

- Facial Recognition: Facial recognition is a technology that is capable of recognizing a person based on their face. It employs machine learning algorithms which find, capture, store and analyze facial features in order to match them with images of individuals in a pre-existing database.
- Image Recognition: Image recognition is one of the most common applications of machine learning. It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, Automatic friend tagging suggestion.
- Autonomous Driving: One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars. Tesla, the most popular car manufacturing company is working on self-driving car. It is using unsupervised learning method to train the car models to detect people and objects while driving.
- Optical Character Recognition: OCR is used to recognize printed text in paper documents, handwritten characters, and text elements in the physical environment, such as license plate numbers, street signs, and street numbers. It is a method that helps

machines recognize texts. Traditional OCR uses patterns and correlation to differentiate words from other elements.

- Speech Recognition: While using Google, we get an option of "Search by voice," it comes under speech recognition, and it's a popular application of machine learning. Speech recognition is a process of converting voice instructions into text, and it is also known as "Speech to text", or "Computer speech recognition." At present, machine learning algorithms are widely used by various applications of speech recognition. Google assistant, Siri, Cortana, and Alexa are using speech recognition technology to follow the voice instructions.

1.3 Machine Learning Libraries

- OpenCV: OpenCV is a highly optimized library with focus on real-time applications. It is a Cross-Platforms for C++, Python and Java interfaces and support Linux, MacOS, Windows, iOS, etc. using OpenCV we can develop real-time computer vision applications like face detection and object detection.
- Yolov3: Yolo (You Only Look Once) is an algorithm that utilizes a single convolutional network for object detection and also provides pre-trained modules to detect the objects fast.
- TensorFlow: TensorFlow is an end-to-end open-source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the state-of-the-art in ML and developers easily build and deploy ML powered applications.
- SQLite: SQLite is a server-less database which can used within almost all programming languages including Python. Server-less means there is no need to install a separate server to work with SQLite so we can connect directly with the database.

1.4 Introduction to python in Machine Learning

Python has lately become the backbone of Machine Learning. It is an easy-to-use programming language compared to any other object-oriented language. Python is often used for data mining and data analysis and supports the implementation of a wide range of Machine Learning models and algorithms. Python is famous for its readability, and it also offers platform independence, which means that you can use the same code for any machine without changes. All these make it the perfect language of choice for Machine Learning.

Python is the preferred language for machine learning is its access to a large number of libraries. Since machine learning heavily relies on mathematical optimization, probability and statistics, Python libraries help data scientists perform various studies easily. Python is undoubtedly the best choice for machine learning. It's easy to understand which makes data validation quick and practically error-free. By having access to a widely developed library ecosystem, developers can perform complex tasks without the need for extensive coding. Thanks to its versatility, testing can be performed easily; tests can be run on any platform including Windows, macOS, Linux, Unix among others.

Most ML Engineers prefer the Python language for Machine Learning. Because as ML Engineers, they are responsible for data extraction, data processing, data refining, and understanding the data to implement in various algorithms. So, they need a programming language that is easy to understand and helps them implement Machine Learning algorithms quickly. They need a language that helps them validate the algorithms instantly, and Python offers all these features. So, they implement projects of ML using Python. Python also has a few more advantages as mentioned below:

- Python has a great library system.
- It has a low-entry barrier.
- Python is flexible and versatile.
- It offers platform independence.
- It has multiple visualization options.
- Python is highly popular.

1.5 Project Introduction

- **Face Recognition**

Human beings have recognition capabilities that are unparalleled in the modern computing era. These are mainly due to the high degree of interconnectivity, adaptive nature, learning skills and generalization capabilities of the nervous system. The human brain has numerous highly interconnected biological neurons which, on some specific tasks, can outperform super computers. A child can accurately identify a face, but for a computer it is a cumbersome task. Therefore, the main idea is to engineer a system which can emulate what a child can do. Advancements in computing capability over the past few decades have enabled comparable recognition capabilities from such engineered systems quite successfully. Early face recognition algorithms used simple geometric models, but recently the recognition process has now matured into a science of sophisticated mathematical representations and matching processes. Major advancements and initiatives have propelled face recognition technology into the spotlight. Face recognition

technology can be used in wide range of applications. Computers that detect and recognize faces could be applied to a wide variety of practical applications including criminal identification etc. Face detection and recognition is used in many places nowadays, verifying websites hosting images and social networking sites. Face recognition and detection can be achieved using technologies related to computer science. Features extracted from a face are processed and compared with similarly processed faces present in the database. If a face is recognized it is known or the system may show a similar face existing in database else it is unknown. In surveillance system if an unknown face appears more than one time then it is stored in database for further recognition. These steps are very useful in criminal identification. In general, face recognition techniques can be divided into two groups based on the face representation they use appearance-based, which uses holistic texture features and is applied to either whole-face or specific face image and feature-based, which uses geometric facial features (mouth, eyebrows, cheeks etc), and geometric relationships between them. (A few example applications are shown in Fig 1.2.)



Fig 1.2 Biometric application

In Figure 1.2 an important aspect is that such technology should be able to deal with various changes in face images, like rotation, changes in expression. Surprisingly, the mathematical variations between the images of the same face due to illumination and viewing direction are almost always larger than image variations due to changes in face identity. This presents a great challenge to face recognition. At the core, two issues are central to successful face recognition algorithms: First, the choice of features used to represent a face. Since images are subject to changes in viewpoint, illumination, and expression, an effective representation should be able to deal with these possible changes. Secondly, the classification of a new face image using the chosen representation.

- **Biometrics**

Biometrics is used in the process of authentication of a person by verifying or identifying that a user requesting a network resource is who he, she, or it claims to be, and vice versa. It uses the property that a human trait associated with a person itself like structure of data with the incoming data we can verify the identity of a particular person. There are many types of biometric system like detection and recognition, iris recognition etc., these traits are used for human identification in surveillance system, criminal identification, face details etc. By comparing the existing fingerprint recognition.

- **Applications of existing system**

Face recognition systems have gained a great deal of popularity due to the wide range of applications that they have proved to be useful in. Broadly, two main categories for these applications exist: commercial applications and research applications.

From a commercial standpoint, face recognition is practical in security systems for law enforcement situations. It is in places like airports and international borders that the need arises for a face recognition system that identifies individuals. Another application of face recognition is the protection of privacy, obviating the need for exchanging sensitive personal information. Instead, a computer-based face recognition system would provide sufficient identification. For instance, PIN numbers, user ID's, and passwords would be replaced by face recognition in order to unify personal identification. Finally, face recognition systems can be used for entertainment purposes in areas like video games and virtual reality.

In research applications, face recognition has opened the door for research in areas like image and video processing. The approaches used in face recognition are useful in the general area of pattern recognition and data classification. Research has also progressed into the realm of neural networks, where the human nervous system is used as a model to attain higher recognition rates. Lastly, face recognition has paved the way for advances in the field of computer vision. Any research in face recognition is a step forward in autonomous vision-based artificial intelligence.

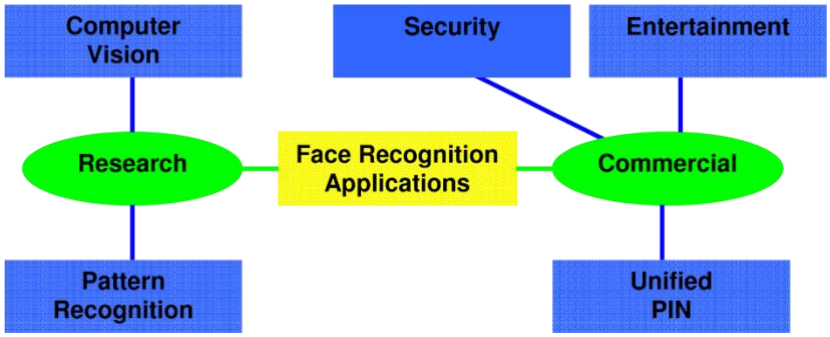


Fig 1.3 Applications of Face Recognition

- **Face Recognition Algorithms**

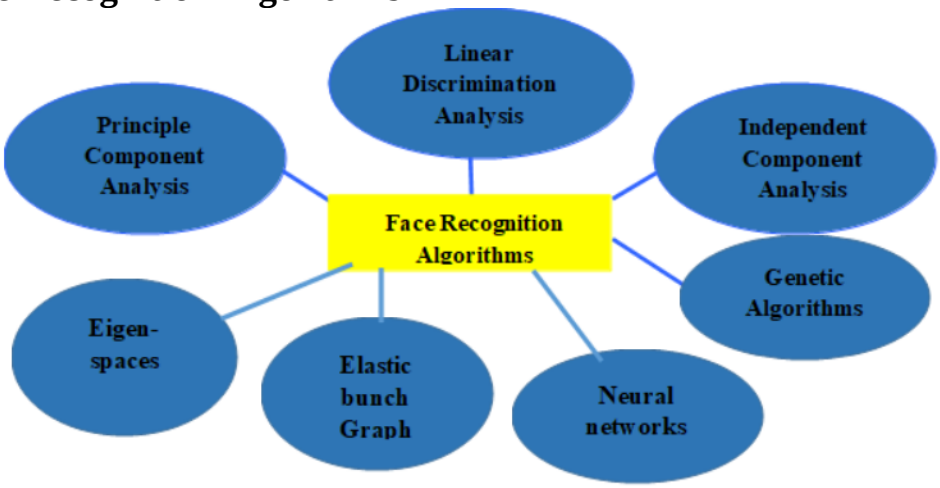


Fig 1.4: Face Recognition Algorithms

1.6 Problem Statement

The problem of face recognition can be stated as follows: Face Recognition human facial features like the mouth, nose and eyes in a full-frontal face image. We will be adapting a multi-step process in order to achieve the goal. To detect the face region, we will be using a skin-color segmentation method. Morphological techniques will be adapted to fill the holes that would be created after the segmentation process. From the skeletonization process, a skeleton of the face will be obtained from which face contour points could be extracted. Facial features can be located in the interior of the face contour. We will use several different facial-images to test our method.

1.7 Objectives of the project

1. Trying to find a face within a large database of faces. In this approach the system returns a possible list of faces from the database. The most useful applications contain crowd surveillance, video content indexing, personal identification (example: driver's license), mug shots matching, etc.
2. Real time face recognition: Here, face recognition is used to identify a person on the spot and grant access to a building or a compound, thus avoiding security hassles. In this case the face is compared against multiple training samples of a person.

- Detection of unique face image amidst the other natural components such as walls, backgrounds etc.
- Extraction of unique characteristic features of a face useful for face recognition.
- Detection of faces amongst other face characters such as beard, spectacles etc.
- Effective recognition of unique faces in a crowd (individual recognition in crowd).
- Automated update in the database without human intervention.

Existing System:

- Fingerprint
- Voice
- Signatures
- Keystroke
- Odour
- Facial, hand, vein infrared thermos gram
- Palm Print
- Hand and Finger Geometry

Motivation:

- Attendance is an important part of daily classroom evaluation.
- At the beginning and ending of class, it is usually checked by the teacher, but it may appear that a teacher may miss someone or some students answer multiple times.

- Face recognition-based attendance system is a problem of recognizing face for taking attendance by using face recognition technology based on high-definition monitor video and other information technology.

Chapter 2 LITERATURE SURVEY

A literature survey in a project is a section that allows various analysis of research made in the field of interest and the result which are already been published by considering various parameters of the project. It is the most important phase of the project as it gives direction for the implementation of the project. The detailed analysis of the methodologies and the techniques used by the researchers should be noted. This chapter describes the key features emphasized by the papers that have been surveyed. Several algorithms and techniques for face recognition have been developed in the past by researchers. These are discussed briefly in this section.

2.1 Facial recognition system for people with and without face mask in times of the COVID-19 Pandemic (2021) by Jonathan S Talahua, Jorge Buele, P Calvopina, Jose Vareela Aldas

The study discusses the many approaches for recognizing people who are wearing masks. The OpenCv face detector is utilized in conjunction with a MobileNetV2 architecture-based classification model. It's easy to detect the face of an individual and whether the person is wearing a mask or not. The FaceNet model is utilized as a feature. A separator and a feedforward multilayer perceptron are used to recognize the features of the face. The coronavirus illness (COVID-19) is an unprecedented disaster which leads to a huge number of deaths and security problems. People regularly wear disguises to defend the counter to the broadcast of COVID-19. Face detection develops extra difficulty owed to the distorting of important facemask features. Experts focus their efforts on the current outbreak of coronavirus in creating quick and traditional remedies. In this study, there is an attempt to address the problem of facial attention by proposing an effective method based on reduction of closure and deep learning-based features. First, remove the mask from the face. To excerpt in-depth structures in the learnt regions, there is use of trio pre-skilled unconventional CNNs: VGG-16, AlexNet, and ResNet-50 (which is mainly eye and forehead regions). When compared to the fully integrated CNN layer, the Bag-of-features paradigm is utilized to map the last feature of the layer conversion to measure and gain less representation. The technology utilized necessitates a considerable quantity of data storage, which will not be available to all users; this is a basic paper oversight.

Author(s): Jonathan S Talahua, Jorge Buele, P Calvopina, Jose Vareela Aldas

Publisher: IEEE Transactions on neural networks

Department: Department of Computer Science and Engineering Date of Conference: 2021

2.2 Masked Face Recognition Using Deep Learning: A Review (2021) by Ahmad Alzu'bi Firas Albalas, Tawfik AL-Hadhrami, Lojin Bani Younis and Amjad Bashayreh

This paper presents contemporary MFR work based on deep learning techniques, as well as insights and a full overview of the MFR system development pipeline. Pandemics and scams, for example, have accelerated the development and distribution of appropriate algorithms, bringing new challenges. As a result, detecting and validating people who wear covers will be a lasting study project, and more effective real-time MFR techniques will be necessary. Machine learning has tremendously enhanced the intelligent process of discovering and validating people with veiled faces, allowing MFR to advance. This assessment examines and evaluates contemporary MFR work that employs deep learning techniques, offering insights and a comprehensive explanation of the MFR system development pipeline. Several brainy models for (MFR) have lately remained labelled and practical in a variety of areas, including masked face trailing for persons security and safe verification. Exceptional threats, pandemics and deceptions have significantly sped production and distribution of pertinent algorithms, posing new obstacles. Low speed is one of the restrictions.

Authors: Ahmad Alzu'bi Firas Albalas, Tawfik AL-Hadhrami, Lojin Bani Younis and Amjad Bashayreh

Publisher: The 5th Asian Conference on Computer Vision, 23-25 January, 2002, Melbourne, Australia

Date of Conference: 2021

2.3 Efficient Masked Face Recognition Method during the COVID-19 Pandemic (2021) by Walid Hariri

To resolve the test of disguise that is mask face recognition, this investigation delivers a dependable method based on obstruction elimination and deep learning-based techniques. People often wear masks to guard themselves from COVID-19 transmission. Face recognition is mainly problematic since central parts of the face are hidden. Researchers are focusing their efforts during the current coronavirus outbreak on creating recommendations for dealing with the problem in a timely and effective manner. In this paper, there is a presentation of a feasible plan based on occlusion elimination and deep learningbased structures to address the trouble of disguised face recognition. The World Health Organization professed the usage of a face masked as a mandatory biosafety amount in the aftermath of the COVID19 epidemic. This has generated issues with current facial recognition systems, prompting the creation of this study. The development of a system for recognising a person from images, even when they wear a face mask, is described in this manuscript. The OpenCv face detector is joint with an organization model founded on the MobileNetV2 construction. As a result of these processes, it is possible to detect where the face is and whether it is wearing a face mask. Face recognition is attained by means of the FaceNet model as a features separator and feedforward multilayer perceptron is used. Deep learning is the methodology used in this paper. Low precision is one of the limitations.

Authors: Walid Hariri

Publisher: Colorado State University

Date of Conference: 2021

2.4 Face Recognition System (2020) by Shivam Singh

This effort presents a request based on face detection, feature extraction of the face, and documentation procedures that notices the humanoid face mechanically when the being in front of the photographic camera recognizes the person's individuality. This program employs face detection, feature extraction, and identification algorithms to detect the human face when the person in front of the camera recognizes him. They have used the KLT approach for feature selection, the Viola-Jones algorithm for face recognition (which employs a Haar cascade classifier to detect human faces but the camera identifies the face every frame), and the PCA technique for feature detection. A model combining approach is used to imitate the geometric characteristics of the human face. The article methodology covers a Face Recognition System of Local and Global. Facial Recognition's dependability might be thrown off by different face angles.

Authors: Shivam Singh

Publisher: Research Gate

Date of Conference: 2020

2.5 A review of face recognition technology (2020) by Lixiang Li,Xiaohuimu, SiyingLi, Haipeng

Face Recognition knowledge is a biometric technology that is founded on the credit of a person's face characters. Introduces related face recognition studies from several perspectives. The study examines relevant face recognition studies from a variety of perspectives. The presentation goes over the many stages of facial recognition as well as related technologies. They also communicated about real-world face recognition education, as well as overall assessment values and face recognition records. PCA is the method employed in this paper. Poor lighting and varying face angles are the drawbacks.

Authors: Lixiang Li,Xiaohuimu, SiyingLi, Haipeng

Publisher: Journal of electrical and computer engineering

Date of Conference: 2020

2.6 Concept of Face recognition (2020) by Dhara Rathore, Gaurav Kumar Das, Dr Akash Saxena

This study examines significant Face Recognition in the current era. Face recognition is a task that has recently received a lot of attention. There is a fundamental need for effective security with so much information and data being collected. Face recognition has been a rapidly emerging, tested, and intriguing topic in terms of current applications. The present state of human facial recognition research is summarized in this report. Model-based procedures, holistic based approaches, support vector machines, and artificial neural networks are some of the FR models used in the study that are cited in this work. The paper's methodology is model-based methods. Massive data application is the restriction.

Authors: Dhara Rathore, Gaurav Kumar Das, Dr Akash Saxena

Publisher: Applied Sciences

Date of Conference: 2020

2.7 Smart Door / COVID-19 Face Mask Detection (2021) by Pavan Narayana A, Janardhan Guptha S

A terrible virus peered into the life of a young lady on January 27, 2020, a day that would be remembered by the Indian people for decades, and it has been so menacing that it has taken the lives of 3.26 lakh people just in India. Since the outbreak of the virus, the government has made it essential to wear masks in crowded or public places such as marketplaces, malls, and private gatherings. As a result, it will be difficult for someone at the entrance to check whether everyone is wearing a mask. In this paper, there is the creation of a smart door face mask recognition system to check whether or not people are wearing masks.

Authors: - Pavan Narayana A, Janardhan Guptha S

Publisher: IEEE

Date of Conference: 2021

2.8 Face Mask Detection Using Deep Learning (2021) by Ravi Kishore Kodali; Rekha Dhanekula

Millions of people have died as a result of a mystery virus that has wrecked devastation on the global landscape. The coronavirus was first detected in late December 2019 in Wuhan, China, according to the World Health Organization (WHO). The virus was found as pathogenic after extensive investigation, and it is spread by the air or direct contact with an infected person. Many strategies for preventing the spread of the virus have been proposed, including keeping a safe distance between persons and avoiding close contact, and wearing a face mask to prevent droplets from circulating in the air. As a result, the goal of this research is to develop and test a system for detecting social distancing and face masks.

Authors: - Ravi Kishore Kodali; Rekha Dhanekula

Publisher: IEEE

Date of Conference: 2021

2.9 Social Distancing and Face Mask Detection using Deep Learning and Computer Vision (2020) by Aldila, D., Khoshnaw, S., Safitri, Erma

A new virus has introduced a global outbreak that has caused massive deaths. According to the World Health Organization, this coronavirus was introduced in late month December 2019 in Wuhan, China (WHO). According to in-depth studies, the virus is pathogenic and is transmitted through the air or through close contact with an infected person. Upholding a communal coldness, or having a good bodily contact among persons and avoiding nearby interaction, as well as tiring a face cover to prevent dewdrops from spreading into the fresh air, have all been cited as ways to reduce the spread of the virus. As a result, the purpose and objective of the investigation article is to create and tool a Communal Distancing and Face Mask Detection System. Thing discovery and facemask identification determination is applied in pedestrian audio-visual film using this system. The YOLOv3, ResNet Classifier, and DSFD are samples of trained replicas. Faces deprived of expression coverings were spotted as well as people violating the minimum distance. The number of persons breaching or not sacrilegious the respective measures is displayed on an overall results board in the output.

Authors: - Aldila, D., Khoshnaw, S., Safitri, Erma

Publisher: IEEE

Date of Conference: 2020

2.10 Face Mask Detection by using Optimistic Convolutional Neural Network (2021) by K Suresh

COVID-19 has wreaked havoc on daily life and has created global health concerns. One of the reasons for survival advice is to wear a protective face mask to avoid coronavirus transmission. The best approach to avoid COVID-19 is to use a face mask. If people aren't wearing their masks properly, it's tough to physically verify and warn the victim in crowded situations. This research proposes a simplified method for detecting and informing individuals who are not wearing a facemask. The suggested system/model is trained and tested using Kaggle datasets. The technology operates in real-time and identifies whether a person's face is covered by a facemask. If it isn't, the person is notified individually via text message. The mask is created using real-time public faces and placed into a convolutional neural network as an input (CNN).

Authors: - K Suresh

Publisher: IEEE

Date of Conference: 2021

2.11 The Face Mask Detection for Preventing the Spread of COVID-19 at Politeknik Negeri Batam (2020) by Susanto Susanto, Febri Alwan Putra

Following a quick epidemic of a novel Coronavirus infection (COVID-19) case in Wuhan, China in the month of December 2019, the World Health Organisation found that this was a risky disease that may have spread between persons via dewdrops and airborne transmission. Before it comes to prevention, putting on a face mask when going outside or meeting new people is a requirement. For a number of reasons, some risky individuals refuse to wear a face mask. Establishing a face cover sensor is also vital in this circumstance. The YOLO V4 profound knowledge mask recognition technique was employed to recognise the facial mask. At Politeknik Negeri Batam, the equipment was installed, and the results of the experiments were used in real time. This technology can accurately identify people who remain wearing or are not wearing a face mask, even if they move around, according to the findings of the tests.

Authors: - Susanto Susanto, Febri Alwan Putra

Publisher: IEEE

Date of Conference: 2020

Table 2.1 Literature Survey

TITLE	AUTHORS	SOURCE	FINDINGS
Facial recognition system for people with and without face mask in times of the COVID-19 Pandemic (2021)	Jonathan S Talahua, Jorge Buele, P Calvopina, Jose Vareela Aldas	IJRASET	Describes the methods of FR for the people wearing mask. CNN have been used as a priority. This tech requires powerful data storage which will not be available to all users.
Masked Face Recognition Using Deep Learning: A Review (2021)	Ahmad Alzu'bi Firas Albalas, Tawfik AL-Hadhrami, Lojin Bani Younis and Amjad Bashayreh	ABMCPL	The recent works developed for MFR based on deep learning techniques, providing insights and thorough discussion on the development pipeline of MFR systems. Deep learning is used. Lower speed.
Efficient Masked Face Recognition Method during the COVID-19 Pandemic (2021)	Walid Hariri	Towards Dat a Science	Propose a reliable method based on occlusion removal and deep learning-based features in order to address the problem of the masked face recognition process. Deep learning is used. Low accuracy.

Face Recognition System (2020)	Shivam Singh	Tyro Labs	This application based on face detection, feature extraction and recognition algorithms which automatically detects the human face when the person in front of the camera recognizing the person's identity. Local Face Recognition System, Global Face Recognition. Different Face Angles Can Throw Off Facial Recognition's Reliability.
A review of face recognition technology (2020)	Lixiang Li,Xiaohuimu, SiyingLi, Haipeng	ResearchGate	Face Recognition technology is biometric technology which is based on the identification of facial features of person. Introduces related researches of face recognition from different perspectives. Uses PCA. Poor lightening and different face angles.
Concept of Face recognition (2020)	Dhara Rathore, Gaurav Kumar Das, Dr Akash Saxena	Applied Sciences	This paper gives a modern audit of significant FR look into. It uses model-based procedures. Massive data application.

Smart Door / COVID-19 Face Mask Detection (2021)	Pavan Narayana A, Janardhan Guptha S	IEEE	There is the creation of a smart door face mask recognition system to check whether or not people are wearing masks.
Face Mask Detection Using Deep Learning (2021)	Ravi Kishore Kodali; Rekha Dhanekula	IEEE	The goal of this research is to develop and test a system for detecting social distancing and face masks.
Social Distancing and Face Mask Detection using Deep Learning and Computer Vision (2020)	Aldila, D., Khoshnaw, S., Safitri, Erma	Springer	the purpose and objective of the investigation article is to create and tool a Communal Distancing and Face Mask Detection System. Thing discovery and facemask identification determination is applied in pedestrian audio-visual film using this system. The YOLOv3, ResNet Classifier, and DSFD are samples of trained replicas.
Face Mask Detection by using Optimistic Convolutional Neural Network (2021)	K Suresh	IEEE	This research proposes a simplified method for detecting and informing individuals who are not wearing a facemask. The suggested system/model is trained and tested using Kaggle datasets. The technology operates in real-time and identifies whether a

			person's face is covered by a facemask.
The Face Mask Detection for Preventing the Spread of COVID-19 at Politeknik Negeri Batam (2020)	Susanto Susanto, Febri Alwan Putra	IEEE	This technology can accurately identify people who remain wearing or are not wearing a face mask, even if they move around, according to the findings of the tests.

Chapter 3 DESIGN AND METHODOLOGY

3.1 Approach

The total system is divided into 3 modules- Database creation, Training the dataset, Testing, sending alert messages as an extension.

1. Database creation

a) Initialize the camera and set an alert message to grab the attention of the students.

b) Get user id as input

c) convert the image into gray scale, detect the face and

d) Store it in database by using given input as label up to 20 frames.

2. Training

a) Initialize LBPH face recognizer.

b) Get faces and Id's from database folder to train the LBPH face recognizer.

c) Save the trained data as xml or yml file.

3. Testing

Load Haar classifier, LBPH face recognizer and trained data from xml or yml file.

a) Capture the image from camera,

b) Convert it into gray scale,

c) Detect the face in it and

d) Predict the face using the above recognizer.

This proposed system uses Viola Jones algorithm for face detection which uses modified Haar Cascades for detection. Raspberry Pi is the main component in the project. We will be using USB webcam to capture photos. We can access Raspberry Pi's console either by using SSH in laptop or by using Keyboard and mouse with the display device like TV
connected to Pi. Firstly, the algorithm needs a lot of positive images and negative images to train the Haar cascades classifier. Positive images are images with clear faces where negative images are those without any faces.

Haar Cascades:

Each feature is represented as a single value obtained from the difference of the sums of pixels in white rectangle from the sum of all pixels in the black rectangle. All different possible sizes and locations of classifier is used for calculating of plenty of features. As the number of classifiers increase the arithmetic computations seems to take a long time. To avoid this, we use the concept

of Integral Image. In Image Processing Integral image is a data structure which is summed area table and algorithm for quickly and efficiently generating sum of values in a rectangular grid subset. Integral image is derived by using the formula.

Integral image:

To solve the complexity of the number of classifiers applied for calculation we use Ad boost machine learning algorithm, which is inbuilt in OpenCV library that is cascade classifier, to eliminate the redundancy of the classifiers. Any classifier which has a probability of 50% of more in detection is treated as weak classifier. The Sum of all weak classifier gives a strong classifier which makes the decision about detection. Although it is very vague to classify with one strong classifier, we use the cascade of classifiers. Classification takes place in stages, if the selected region fails in the first stage, we discard it. We don't use the classifiers on that region which is discarded. The region which passes all the stages i.e., all strong classifiers is treated as the detected face. Detected Faces are passed to the Face recognition phase. In this phase we use Local Binary Patterns algorithm for face recognition. Local binary patterns are simple at the same time very efficient texture operator which assigns the pixels of the image by comparing with the adjacent pixels as threshold and which results in a binary result. The detected integral image is subjected to this Local binary pattern which results in decimals are represented as histogram for every integral image. Face recognition is extremely vulnerable to the environment changes like brightness, facial expressions and position. Face pre-processing is the module which reduces the problems that makes the picture unclear to recognize the face such as less brightness and contrast problems and noise in the image and make sure the facial features always be in a constant position. In this project we use histogram equalization for face pre-processing. For efficiency we use separate pre-processing which is histogram equalization for left and right face. So, histogram equalization is done three times, firstly for the whole face and the other two for side faces.

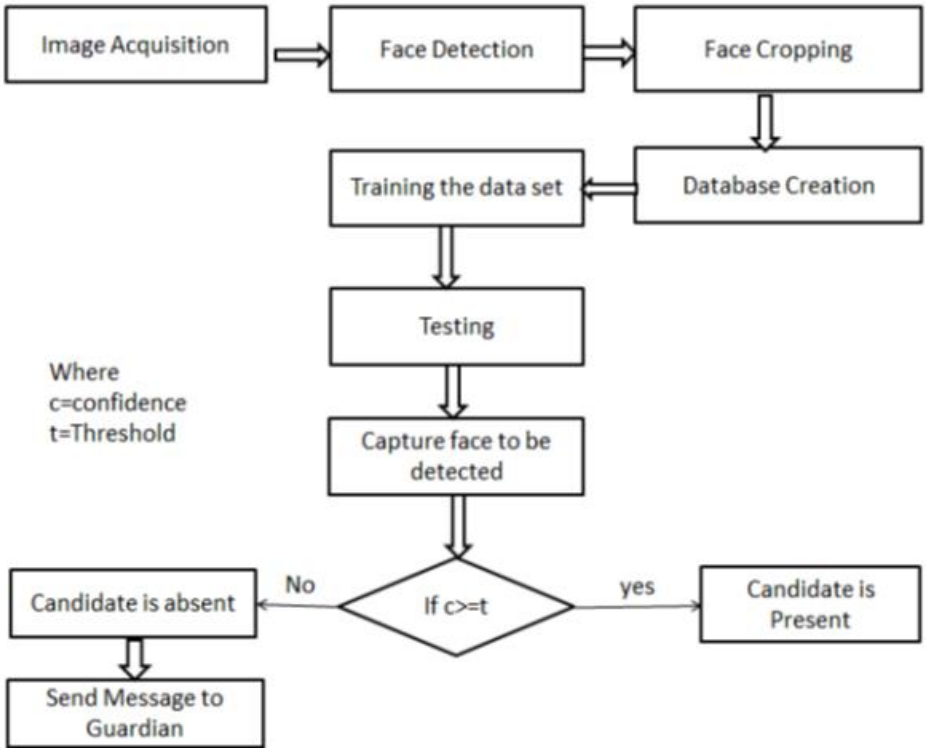


Fig 3.1 Approach to the system

3.2 System Design

System design is the process of the defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. Systems design could be seen as the application of systems theory to product development. Object- oriented analysis and methods are becoming the most widely used methods for computer systems design. Systems design is therefore the process of defining and developing systems to satisfy specified requirements of the user. The UML has become the standard language in object-oriented analysis and design. System architecture is a conceptual model that defines the structure and behaviour of the system. It comprises of the system components and the relationship describing how they work together to implement the overall system.

3.3 System Architecture

A system architecture is the conceptual model that defines the structure, behavior, and more views of a system. An architecture description is a formal description and representation of a system, organized in a way that supports reasoning about the structures and behaviors of the system. A system architecture can consist of system components and the sub-systems developed which will work together to implement the overall system. There have been improvements in the field of Architecture Description Languages. Architecture description languages are formal languages which are used to describe the system architecture.

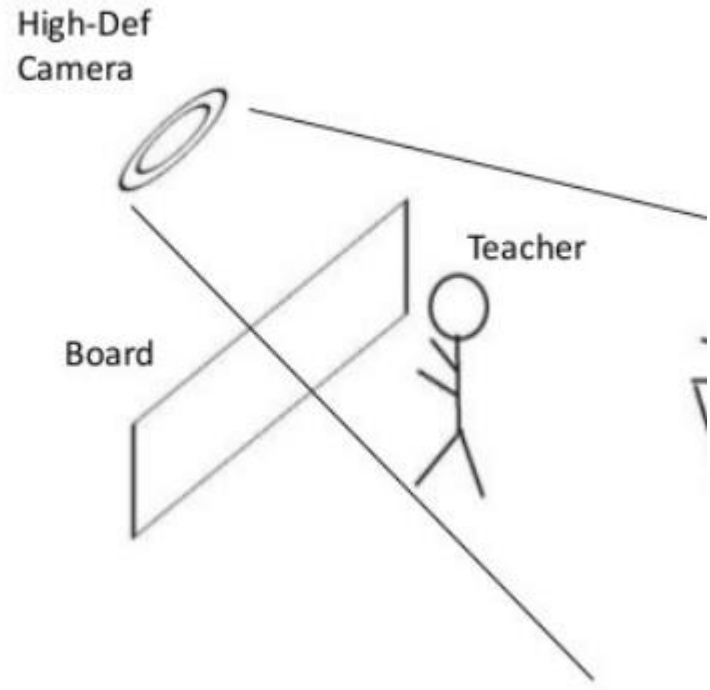


Fig 3.2 System architecture overview

Sequence Diagram

A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are typically associated with use case realizations in the Logical View of the system under development.

Sequence diagrams are sometimes called event diagrams or event scenarios. A sequence diagram shows, as parallel vertical lines (lifelines), different processes or objects that live simultaneously,

and, as horizontal arrows, the messages exchanged between them, in the order in which they occur. This allows the specification of simple runtime scenarios in a graphical manner.

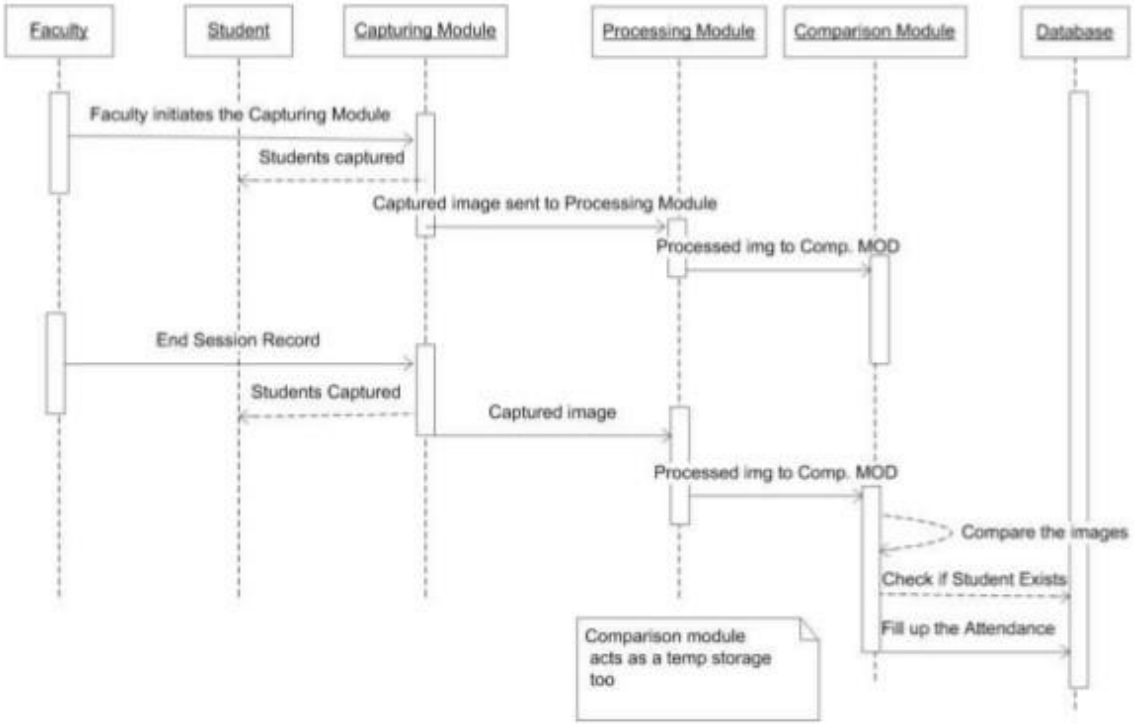


Fig 3.3 Sequence Diagram

3.4 System Requirement Analysis

DEEP LEARNING

1) Photo processing: There are a few sets of data available that include pairing of face and off-face masks to adequately train an FR system in a continuous way.

2) In-depth Learning Models: In-depth learning methods were developed rapidly and applied to many in-depth layers of feature rendering and image modification. Conventional Neural Networks CNNs often incorporate cascade coatings to switch the rate of change, ruler and alteration, namely effort, official, small sample, completely connected, and production layers.

3) Feature Removal: Feature removal is an important stage in facial tube that aims to remove a set of discriminating features enough to represent and study key facial features such as facial features.

4) Mask Detection: Most of the available diagnostic methods, which are usually presented for detection, are filtered and investigated in the mask-finding process.

5) Look Disclosure: The numerous methods accepted in the elimination manual, is a mask in this research context. Several common methods in this research setting. A few common methods are introduced here to remove something based on learning and deleted algorithms that are not based on learning.

6) Face Recovery: after face retrieval, any missing parts must be weighed and returned for further identification in order to make a recognition decision.

7) Face Matching and Recognition: Face corresponding with deep structures of Face Recognition and Masked Face Recognition can be careful a problematic for facial confirmation or documentation. To accomplish this task, a set of photographs of the identified topic was given at the beginning of the program throughout the exercise and verification stage. In the test stage, a new invisible topic which is introduced to the organization to make a gratitude choice.

First, a group of original hidden imageries with consistent low-key imageries is edited. typically involves categorizing in to categories for the drive of perfect exercise, authentication, and assessment. This remains shadowed by other pre-capture functions like adding information and copy division. Such structures must be biased plenty to gain a facial mask precisely. The facial removal procedure is then used to restore the covered face and restore the true facial size. Finally, a predicted face is compared to a real face down to determine if someone has been identified. The effectiveness of Face Recognition programs, by or lacking masks, is greatly prejudiced through the type of facial imageries castoff in the exercise, verification, and evaluation phases. There are a number of openly obtainable databases which includes a pair of face and off-face masks to adequately trains the Masked Face Recognition system in a continuous means. So, reinforces the need to enrich the test-bed with extra artificial imageries with a variety of face masks and to develop the ability to integrate in-depth knowledge replicas. Amongst the greatest general approaches castoff to combine face masks are Mask-The-Face, Masked FaceNet deep Constitutional Neural Network (DCNN), CYCLE-GAN Identity Aware Mask GAN (IAMGAN), and starGAN. Images were re-processed in advance using augmentation of data, where many functions can be used to augment the value and difference of imageries like collecting, tossing, rotating, and aligning. Additional enhancement techniques remain too used to recover image representation excellence such as copy resizing, splitting, audio removal, or sliding.

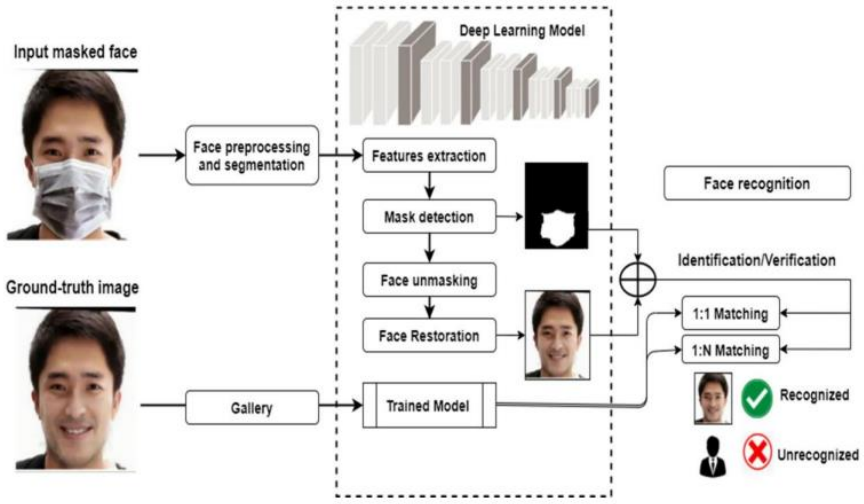


Fig 3.4: A pictorial representation of the masked face recognition framework.

DIGITAL IMAGE PROCESSING

Interest in digital image processing methods stems from two principal application areas:

- Improvement of pictorial information for human interpretation
- Processing of scene data for autonomous machine perception in this second application area, interest focuses on procedures for extracting image information in a form suitable for computer processing. Examples includes automatic character recognition, industrial machine vision for product assembly and inspection, military recognizance, automatic processing of fingerprints etc.

IMAGE

An image refers a 2D light intensity function f (x, y), where (x, y) denotes spatial coordinates and the value of f at any point (x, y) is proportional to the brightness or gray levels of the image at that point. A digital image is an image f (x, y) that has been discretized both in spatial coordinates and brightness. The elements of such a digital array are called image elements or pixels.

A SIMPLE IMAGE MODELS

To be suitable for computer processing, an image f (x, y) must be digitalized both spatially and in amplitude. Digitization of the spatial coordinates (x, y) is called image sampling. Amplitude digitization is called gray-level quantization.

The storage and processing requirements increase rapidly with the spatial resolution and the number of gray levels.

Example: A 256 gray-level image of size 256x256 occupies 64k bytes of memory.

Types of image processing

- Low level processing
- Medium level processing
- High level processing

Low level processing means performing basic operations on images such as reading an image resize, resize, image rotate, RGB to gray level conversion, histogram equalization etc…, The output image obtained after low level processing is raw image. Medium level processing means extracting regions of interest from output of low-level processed image. Medium level processing deals with identification of boundaries i.e., edges. This process is called segmentation. High level processing deals with adding of artificial intelligence to medium level processed signal.

FUNDAMENTAL STEPS IN IMAGE PROCESSING

Fundamental steps in image processing are

1. Image acquisition: to acquire a digital image
2. Image pre-processing: to improve the image in ways that increases the chances for success of the other processes.
3. Image segmentation: to partitions an input image into its constituent parts of objects.
4. Image segmentation: to convert the input data to a from suitable for computer processing.
5. Image description: to extract the features that result in some quantitative information of interest of features that are basic for differentiating one class of objects from another.
6. Image recognition: to assign a label to an object based on the information provided by its description.

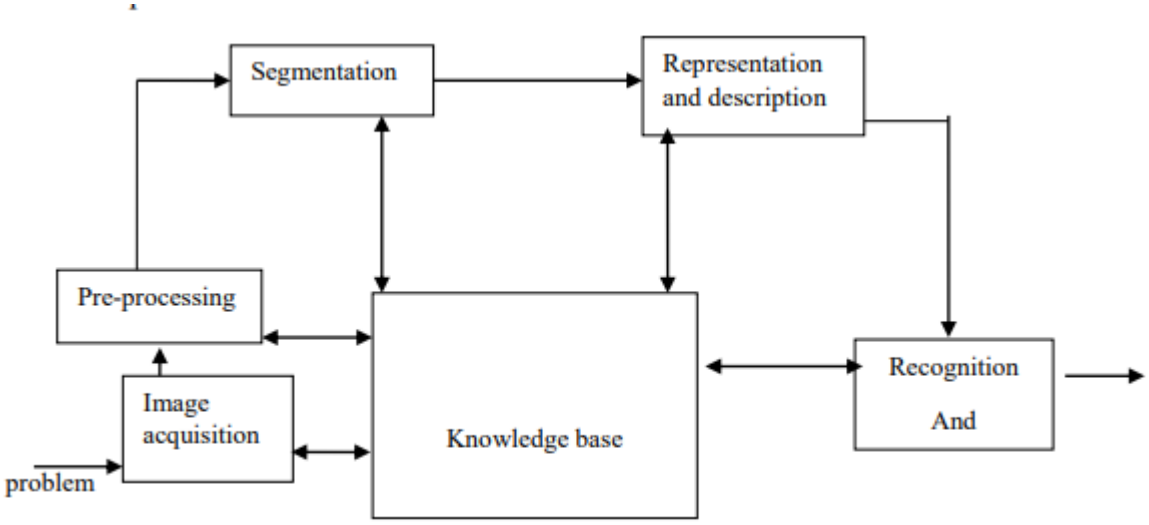


Fig 3.5: Digital Image Processing Steps

ELEMENTS OF DIGITAL IMAGE PROCESSING SYSTEMS

A digital image processing system contains the following blocks as shown in the figure.

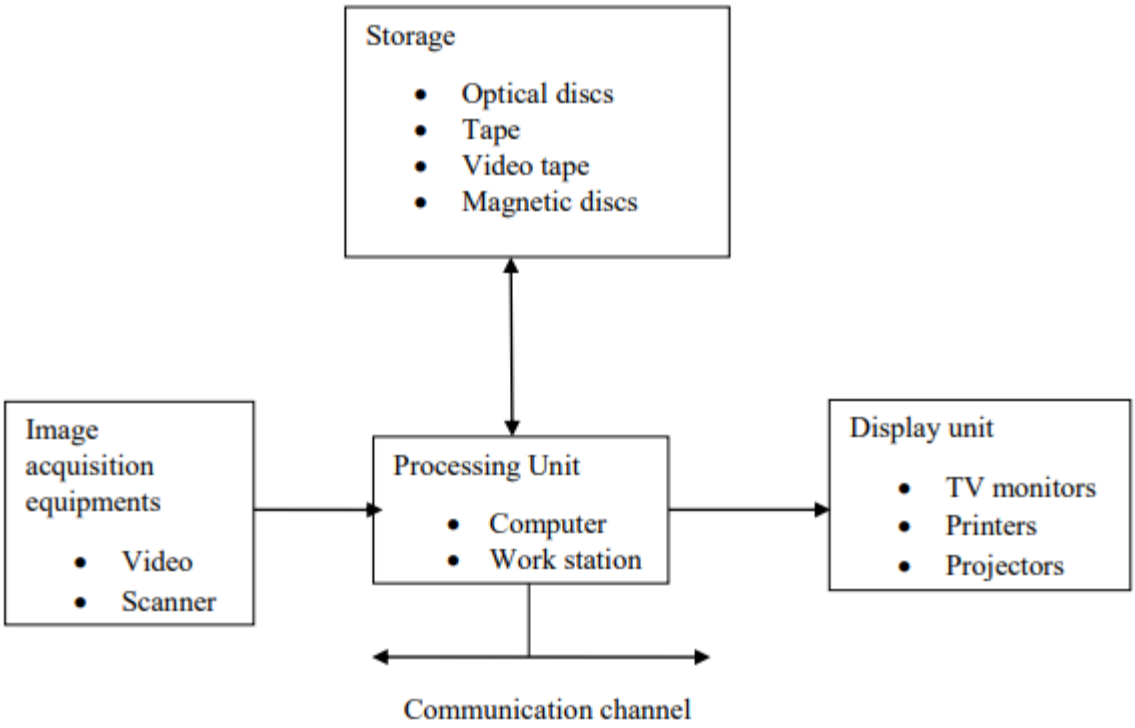


Fig 3.6: Elements of digital image processing

Elements of digital image processing systems the basic operations performed in a digital image processing system include

Acquisition

Storage

Processing

Communication

Display

A SIMPLE IMAGE FORMATION MODEL

Image is denoted by two-dimensional function f (x, y). f (x, y) may be characterized by components:

1. The amount of source illumination i(x, y) incident on the scene

2. The amount of illumination reflected r(x, y) by the objects of the scene

3. f (x, y) = i(x, y) r (x, y), where 0 < i(x,y) < and 0 < r (x, y) < 1

Typical values of reflectance r (x, y):

• 0.01 for black velvet

• 0.65 for stainless steel

• 0.8 for flat white wall paint

• 0.9 for silver-plated metal

• 0.93 for snow Example of typical ranges of illumination i(x, y) for visible light (average values)

• Sun on a clear day: ~90,000 lm/m^2, down to 10,000lm/m^2 on a cloudy day

• Full moon on a clear evening: -0.1 lm/m^2

• Typical illumination level in a commercial office. ~1000lm/m^2 image Formats (supported by open cv Image Processing Toolbox).

3.5 Algorithms

3.5.1 Haar Cascade

Haar Cascade classifiers are an effective way for object detection. This method was proposed by Paul Viola and Michael Jones in their paper Rapid Object Detection using a Boosted Cascade of Simple Features . Haar Cascade is a machine learning-based approach where a lot of positive and negative images are used to train the classifier.

- Positive images – These images contain the images which we want our classifier to identify.
- Negative Images – Images of everything else, which do not contain the object we want to detect.

3.5.2 LBPH

Local Binary Patterns Histogram algorithm was proposed in 2006. It is based on local binary operator. It is widely used in facial recognition due to its computational simplicity and discriminative power.

The steps involved to achieve this are:

- creating dataset
- face acquisition
- feature extraction

- classification

The LBPH algorithm is a part of OpenCV.

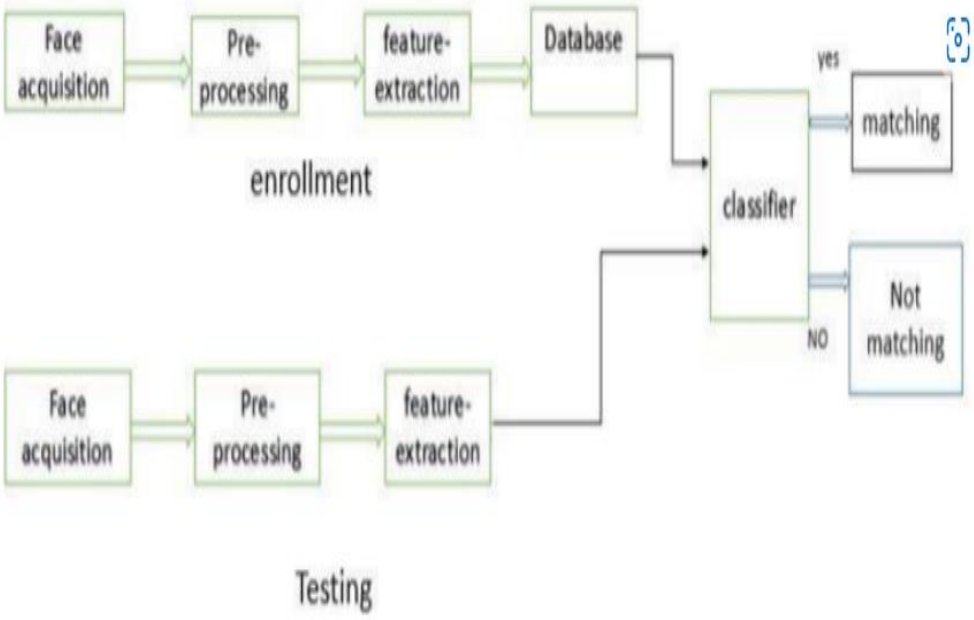


Fig 3.7: Steps in LBPH Algorithm

3.5.3 CNN

The CNN (Convolutional Neural Network) remains a deep learning scheme that container income an image of the contribution, allocate the values of many items / objects to an image, and distinguish between them. CNN is a pattern-detection method that is frequently used in photo analysis. The convolutional neural network (CNN) is a kind of neural network that includes one or more lively coats and is used for image dispensation, splitting, and additional errands. CNN is an excellent detection approach for pattern discovery. CNN needs actual slight previous dispensation in difference to other organization approaches. CNN has a structure that resembles the human brain's neuronal transmission network. Separate neurons can only reply to the scheme in a minor helping of the open field. Such fields cover the entire observable area.

The object or image can be recognized using four-types of layers for a CNN:

• ReLU Layer

• Convolutional Layer

• Fully Connected Layer

• Pooling Layer

Convolutional Layout:

CNN's convolution layer is an essential component. This layer's main goal is to perceive the existence of a collection of features in imageries obtained from data sets as inputs. The feature is

represented as a filter, and the purpose is to strain the window signifying the feature to the portrayal and compute the convolution produce amid the element and every component of the skimmed image. The layer recognizes few photos as being included and calculates each filter's specific conversions. Strainer's game the features we're observing for in the pictures.

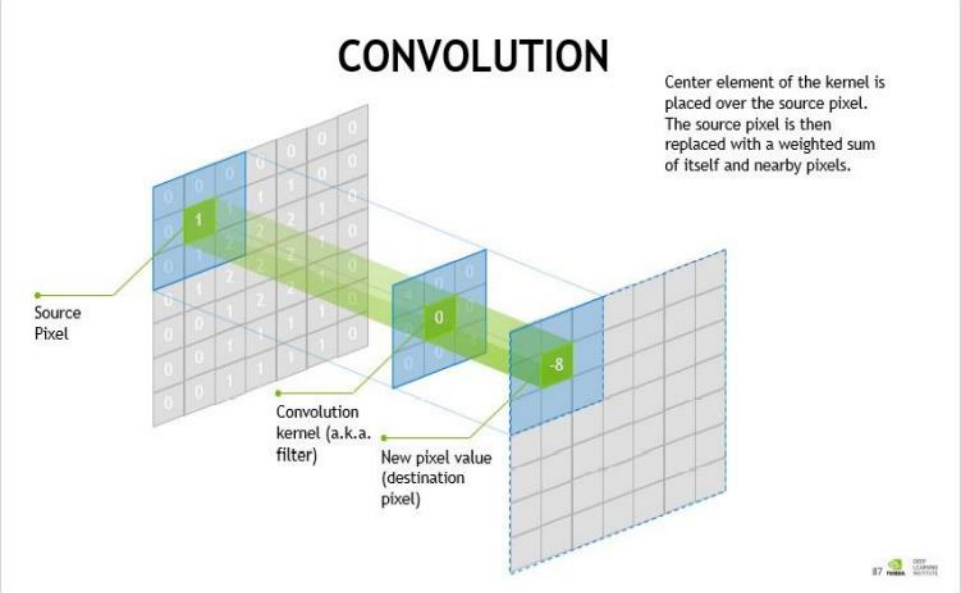


Fig 3.8: A pictorial representation of the convolutional layer

ReLU Correction Layout:

The indirect function defined by ReLU (x) = magnitude is referred to as ReLU (Adjusted Line Units) (0, x). This coating strainers out all negative standards and alternates them with nothing. It also has the skill to function as an activator.

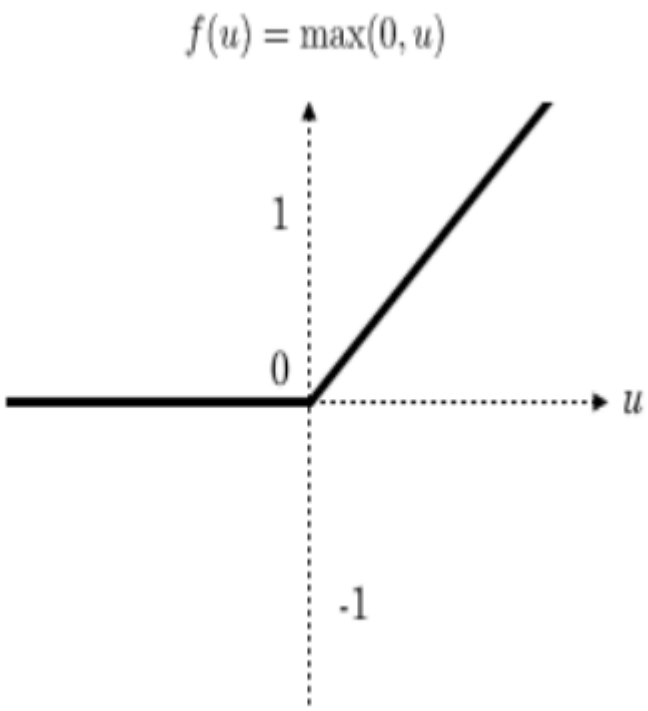


Fig 3.9: Graphic picture of ReLU layer

Pooling Layer:

Among two convolutional layers, a pooling layer is often used. It takes a number of features charts and smears the pooling process to apiece one. This layer's primary function is to reduce the size of photos while maintaining their essential qualities. To carry out, divide the image into uniform cells, keeping the highest rate in every cell. The most popular options are 2*2 adjacent cells that don't overlap and 3*3 cells that are separated by a 2-pixel step and so overlap. The number of feature maps produced is the same as the contribution, but they are substantially compact. The poling cover minimizes the number of computations and parameters in the network, resulting in increased network efficiency. Max Pooling Layer and Threshold Layer stand two substitute stages. The Max Pooling layer reduces the image's undesirable elements, such as noise and brightness. The threshold layer helps in the division of contribution pictures.

Fully Connected Layer:

A fully connected layer is the rearmost layer in a neural system, it checks whether it be convolutional or not. This layer takes a contribution course and alters it into a novel production course. It activates the input values obtained using a line combination and maybe the function. The layer separates the image as input into the network, returning to the selected image a vector of scope N and N number of classes. Each vector element represents the likelihood that the inserted image will be found in the classroom. The association amongst the position of the items in the depicted image and the category is determined by a completely integrated layer. The contribution table, as well as the feature chart, are the grades of the preceding layer.

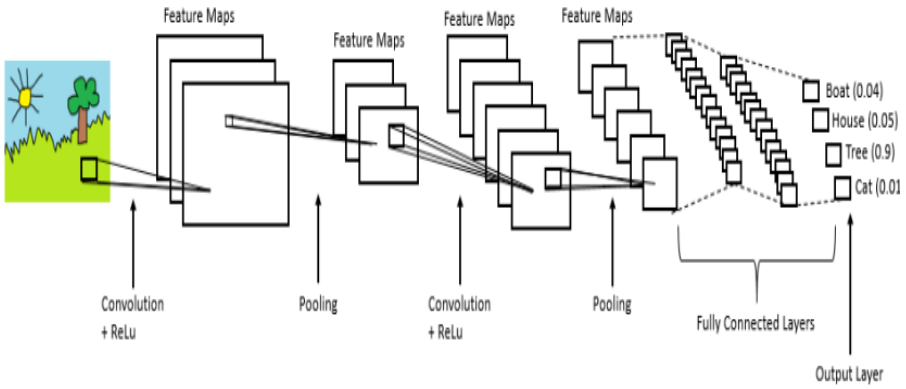


Fig 3.10: Whole CNN construction

Chapter 4

System Requirement Specifications

Hardware Requirements

- Processor: 1GHz or faster
- SD Memory Card: 8GB or above
- Raspberry Pi 4 8GB RAM
- Quantum Hi-Tech USB Camera
- Raspberry Pi 3 HDMI to VGA Cable
- 5V 3A ERD power adapter
- Raspberry Pi screen

Software Requirements

- Operating System: Windows
- Programming Language: Python
- Libraries: TensorFlow, OpenCV

Functional Requirements

- Develop a system in which we can maintain and record the attendance of a lecture and Students
- System Should Automatically capture the images
- System Should Automatically Detect the faces in Image Using Haar Cascade
- System Should Automatically Recognize the faces with the respect to trained faces
- System Should Automatically Update the Database
- System Should Automatically Intimate to Parents

Non-Functional Requirements

- Usability

→ Easy Interface for capture of image

→ Automatically Update the attendance

- Reliability

→ Easy Update

→ Automatic Intimation

- Performance

→ Should not take excessive time in Recognizing the Face

- Supportability

→ Contain easy to understand code with provisions for future enhancement

- Cost

→ The cost of constructing the system should be affordable to all Institutions

Chapter 5 IMPLEMENTATION

5.1 Python IDE

Python is an easy to learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. The Python interpreter is easily extended with new functions and data types implemented in C or C++ (or other languages callable from C). Python is also suitable as an extension language for customizable applications.

5.2 Open CV

OpenCV is a library of programming functions mainly aimed at real-time computer vision. It has a modular structure, which means that the package includes several shared or static libraries. We are using image processing module that includes linear and non-linear image filtering, geometrical image transformations (resize, affine and perspective warping, and generic table-based remapping), color space conversion, histograms, and so on. Our project includes libraries such as Viola-Jones or Haar classifier, LBPH (Lower Binary Pattern histogram) face recognizer, Histogram of oriented gradients (HOG).

5.3 Open CV – Python

Python is a general-purpose programming language started by **Guido van Rossum**, which became very popular in short time mainly because of its simplicity and code readability. It enables the programmer to express his ideas in fewer lines of code without reducing any readability. Compared to other languages like C/C++, Python is slower. But another important feature of Python is that it can be easily extended with C/C++. This feature helps us to write computationally intensive codes in C/C++ and create a Python wrapper for it so that we can use these wrappers as Python modules. This gives us two advantages: first, our code is as fast as original C/C++ code (since it is the actual C++ code working in background) and second, it is very easy to code in Python. This is how OpenCV-Python works, it is a Python wrapper around original C++ implementation. And the support of Numpy makes the task easier. **Numpy** is a highly optimized library for numerical operations. It gives a MATLAB-style syntax. All the OpenCV array structures are converted to-and-from Numpy arrays. So whatever operations you can do in Numpy, you can combine it with OpenCV, which increases number of weapons in your arsenal. Besides that, several other libraries like SciPy, Matplotlib which supports Numpy can be used with this. So OpenCV-Python is an appropriate tool for fast prototyping of computer vision problems.

OpenCV-Python working

OpenCV introduces a new set of tutorials which will guide you through various functions available in OpenCV-Python. **T**his guide is mainly focused on OpenCV 3.x version (although most of the tutorials will work with OpenCV 2.x also).

A prior knowledge on Python and Numpy is required before starting because they won't be covered in this guide. Especially, a good knowledge on Numpy is must to write optimized codes in OpenCV-Python.

OpenCV Needs us.

Since OpenCV is an open-source initiative, all are welcome to make contributions to this library. And it is same for this tutorial also. So, if you find any mistake in this tutorial (whether it be a small spelling mistake or a big error in code or concepts, whatever), feel free to correct it. And that will be a good task for freshers who begin to contribute to open-source projects. Just fork the OpenCV in GitHub, make necessary corrections and send a pull request to OpenCV. OpenCV developers will check your pull request, give you important feedback and once it passes the approval of the reviewer, it will be merged to OpenCV. Then you become an open-source contributor. Similar is the case with other tutorials, documentation etc.

Getting Started with Images

Goals

- Here, you will learn how to read an image, how to display it and how to save it back
- You will learn these functions: **cv2.imread()**, **cv2.imshow(), cv2.imwrite()**
- Optionally, you will learn how to display images with Matplotlib

Using OpenCV

Read an image

Use the function **cv2.imread()** to read an image. The image should be in the working directory or a full path of image should be given.

Second argument is a flag which specifies the way image should be read.

- cv2.IMREAD_COLOR : Loads a color image. Any transparency of image will be neglected. It is the default flag.
- cv2.IMREAD_GRAYSCALE : Loads image in grayscale mode
- cv2.IMREAD_UNCHANGED : Loads image as such including alpha channel

Display an image

Use the function **cv2.imshow()** to display an image in a window. The window automatically fits to the image size.

First argument is a window name which is a string. second argument is our image. You can create as many windows as you wish, but with different window n**cv2.waitKey()** is a keyboard binding function. Its argument is the time in milliseconds. The function waits for specified milliseconds for any keyboard event. If you press any key in that time, the program continues. If **0** is passed, it waits indefinitely for a key stroke. It can also be set to detect specific key strokes like, if key *a* is pressed etc which we will discuss below.

cv2.destroyAllWindows() simply destroys all the windows we created. If you want to destroy any specific window, use the function **cv2.destroyWindow()** where you pass the exact window name as the argument.

5.4 Image Processing Module

Purpose of Image processing

The purpose of image processing is divided into 5 groups. They are:
1. Visualization- Observe the objects that are not visible.
2. Image sharpening and restoration- To create a better image.
3. Image retrieval- Seek for the image of interest.
4. Measurement of pattern– Measures various objects in an image.
5. Image Recognition– Distinguish the objects in an image.

Haar Classifier

This object detection framework is to provide competitive object detection rates in real-time like detection of faces in an image. A human can do this easily, but a computer needs precise instructions and constraints. To make the task more manageable, Viola–Jones requires full view frontal upright

faces. Thus, in order to be detected, the entire face must point towards the camera and should not be tilted to either side. While it seems these constraints could diminish the algorithm's utility somewhat, because the detection step is most often followed by a recognition step, in practice these limits on pose are quite acceptable the characteristics of Viola–Jones algorithm which make it a good detection algorithm are:

a) Robust – very high detection rate (true-positive rate) & very low false-positive rate always.

b) Real time – For practical applications at least 2 frames per second must be processed.

c) Face detection only (not recognition) - The goal is to distinguish faces from non-faces (detection is the first step in the recognition process).

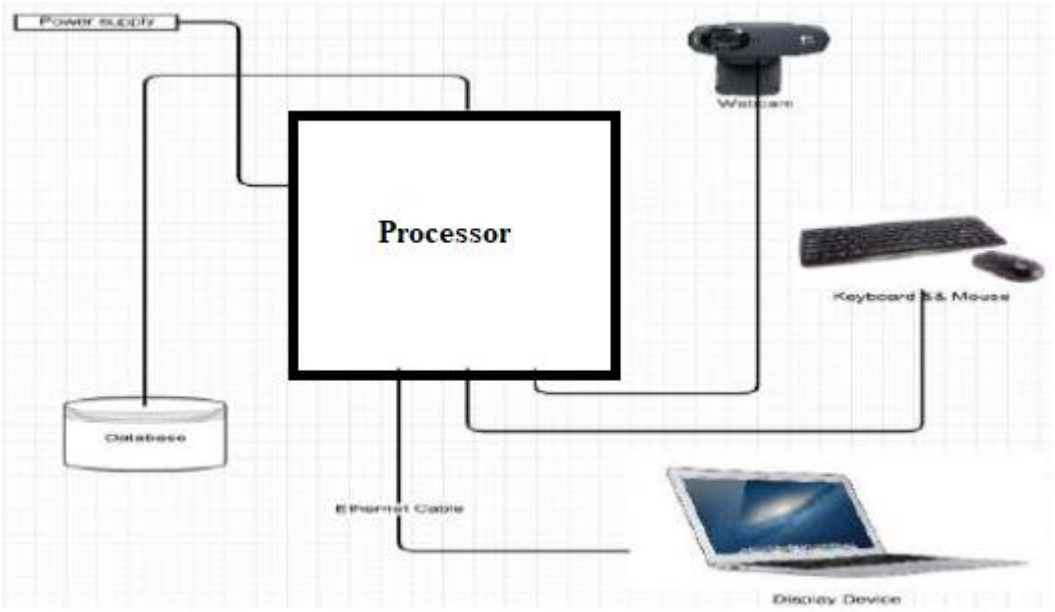


Fig 5.1: Implementation

This algorithm includes Haar feature selection process.

All human faces share some similar properties. These regularities may be matched using Haar Features.

A few properties common to human faces:

a) The eye region is darker than the upper-cheeks.

b) The nose bridge region is brighter than the eyes.

Composition of properties forming matchable facial features:

a) Location and size: eyes, mouth, bridge of nose

b) Value: oriented gradients of pixel intensities

Histogram of oriented gradients (HOG)

Histogram of oriented gradients (HOG) is a feature descriptor used to detect objects in computer vision and image processing. The HOG descriptor technique counts occurrences of gradient orientation in localized portions of an image - detection window, or region of interest (ROI).

Implementation of the HOG descriptor algorithm is as follows:

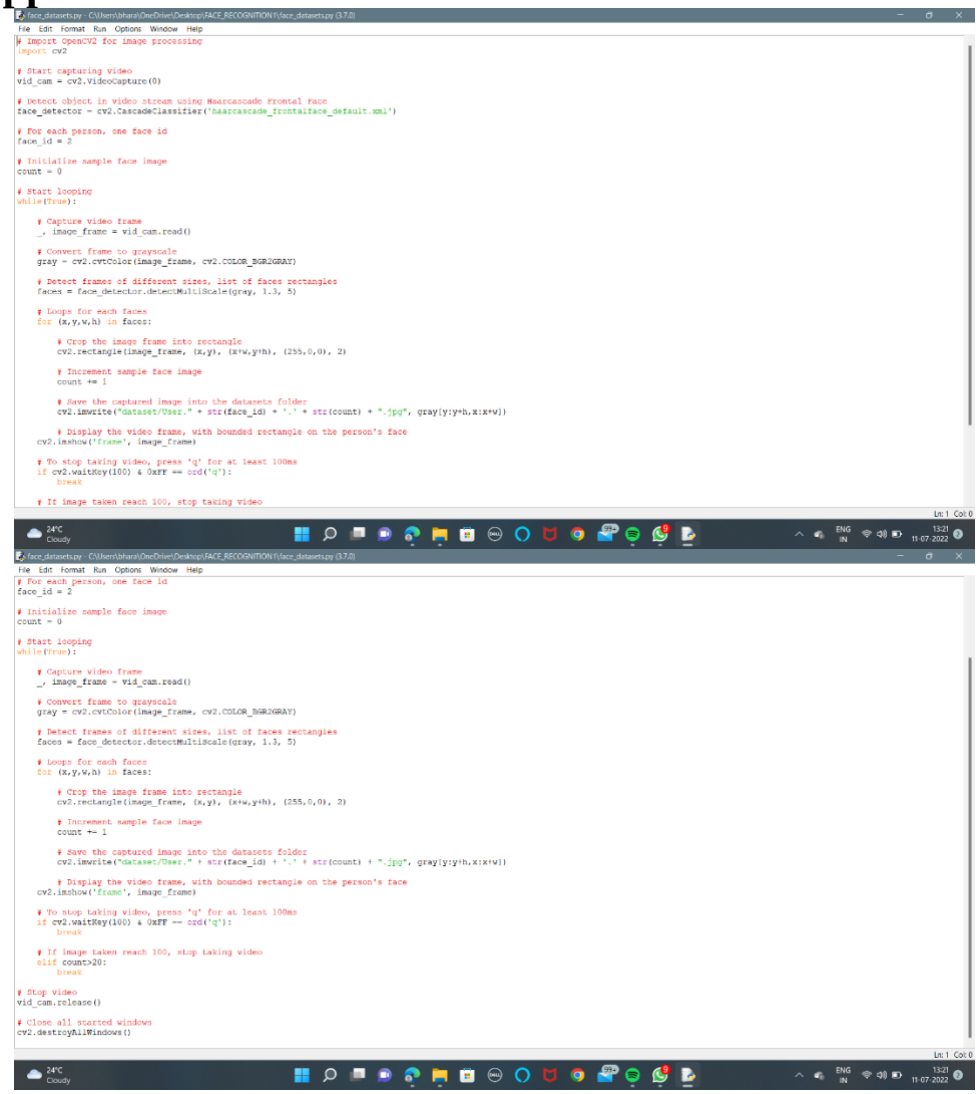
1. Divide the image into small connected regions called cells, and for each cell compute a histogram of gradient directions or edge orientations for the pixels within the cell.
2. Discretize each cell into angular bins according to the gradient orientation.
3. Each cell's pixel contributes weighted gradient to its corresponding angular bin.
4. Groups of adjacent cells are considered as spatial regions called blocks. The grouping of cells into a block is the basis for grouping and normalization of histograms.
5. Normalized group of histograms represents the block histogram. The set of these block histograms represents the descriptor.

5.2 PACKAGES IMPORTED

- Keras: Keras is an open-source software library that provides a Python interface for artificial neural networks. Keras acts as an interface for the tensor flow library. Keras has support for convolutional and recurrent neural networks. It supports other common utility layers like dropout, batch normalization and pooling.
- TensorFlow: It is a foundation library that can be used to create Deep Learning models directly or by using wrapper libraries that simplify the process built on top of TensorFlow.
- NumPy: NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays. It is used to compute scientific calculations.

- OpenCV: OpenCV is a cross-platform library using which we can develop real- time computer vision applications. It mainly focuses on image processing, video capture and analysis including features like face detection and object detection.

5.2 Snippet code for data collection



```
# Import OpenCV2 for image processing
import cv2

# Start capturing video
vid_cam = cv2.VideoCapture(0)

# Detect object in video stream using Haarcascade Frontal Face
face_detector = cv2.CascadeClassifier('haarcascade_frontalface_default.xml')

# For each person, one face id
face_id = 2

# Initialize sample face image
count = 0

# Start looping
while(True):

    # Capture video frame
    _, image_frame = vid_cam.read()

    # Convert frame to grayscale
    gray = cv2.cvtColor(image_frame, cv2.COLOR_BGR2GRAY)

    # Detect frames of different sizes, list of faces rectangles
    faces = face_detector.detectMultiScale(gray, 1.3, 5)

    # Loops for each faces
    for (x,y,w,h) in faces:

        # Crop the image frame into rectangle
        cv2.rectangle(image_frame, (x,y), (x+w,y+h), (255,0,0), 2)

        # Increment sample face image
        count += 1

        # Save the captured image into the datasets folder
        cv2.imwrite("dataset/User." + str(face_id) + '.' + str(count) + ".jpg", gray[y:y+h,x:x+w])

        # Display the video frame, with bounded rectangle on the person's face
    cv2.imshow('frame', image_frame)

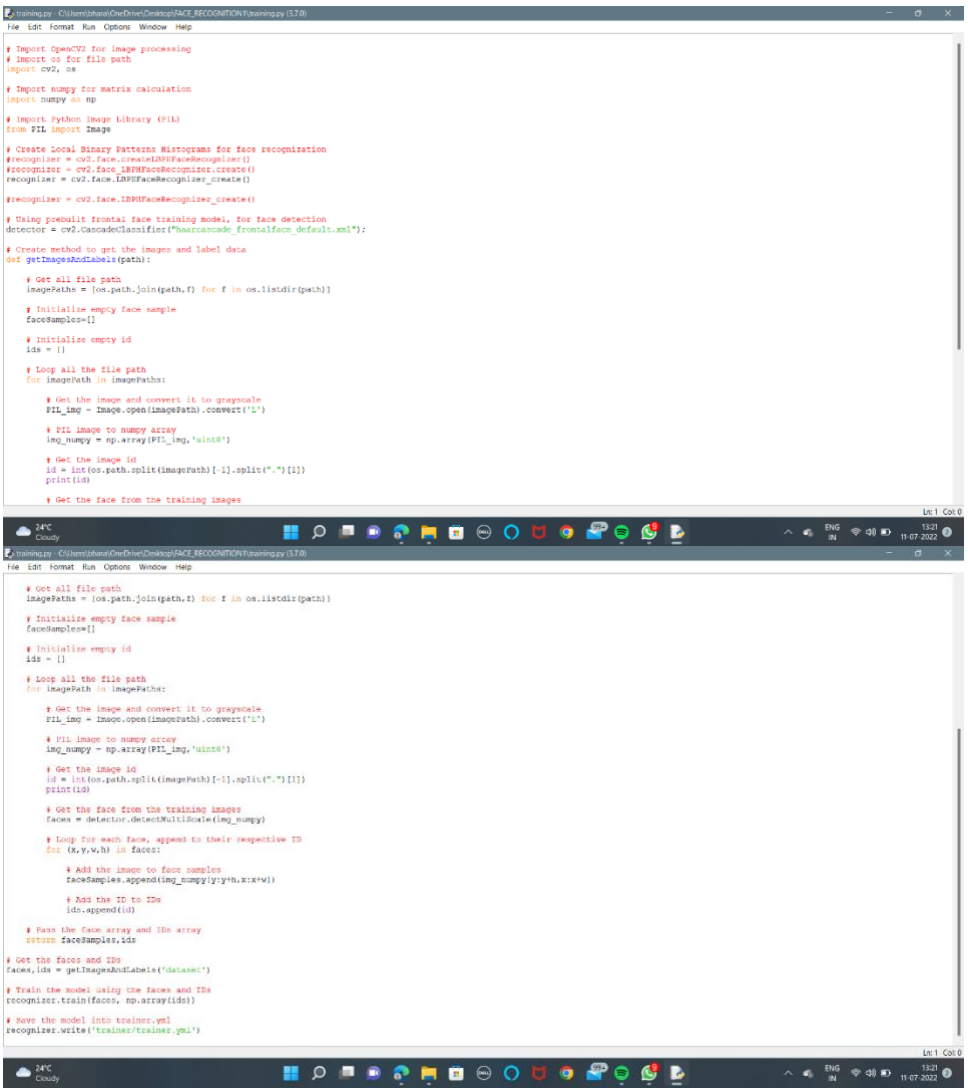
    # To stop taking video, press 'q' for at least 100ms
    if cv2.waitKey(100) & 0xFF == ord('q'):
        break

    # If image taken reach 100, stop taking video
    elif count>20:
        break

# Stop video
vid_cam.release()

# Close all started windows
cv2.destroyAllWindows()
```

5.3 Snippet code for training of data collected



```
# Import OpenCV2 for image processing
# Import os for file path
import cv2, os

# Import numpy for matrix calculation
import numpy as np

# Import Python Image Library (PIL)
from PIL import Image

# Create Local Binary Patterns Histograms for face recognization
#recognizer = cv2.face.createLBPHFaceRecognizer()
#recognizer = cv2.face_LBPHFaceRecognizer.create()
recognizer = cv2.face.LBPHFaceRecognizer_create()

#recognizer = cv2.face.LBPHFaceRecognizer_create()

# Using prebuilt frontal face training model, for face detection
detector = cv2.CascadeClassifier("haarcascade_frontalface_default.xml");

# Create method to get the images and label data
def getImagesAndLabels(path):

    # Get all file path
    imagePaths = [os.path.join(path,f) for f in os.listdir(path)]

    # Initialize empty face sample
    faceSamples=[]

    # Initialize empty id
    ids = []

    # Loop all the file path
    for imagePath in imagePaths:

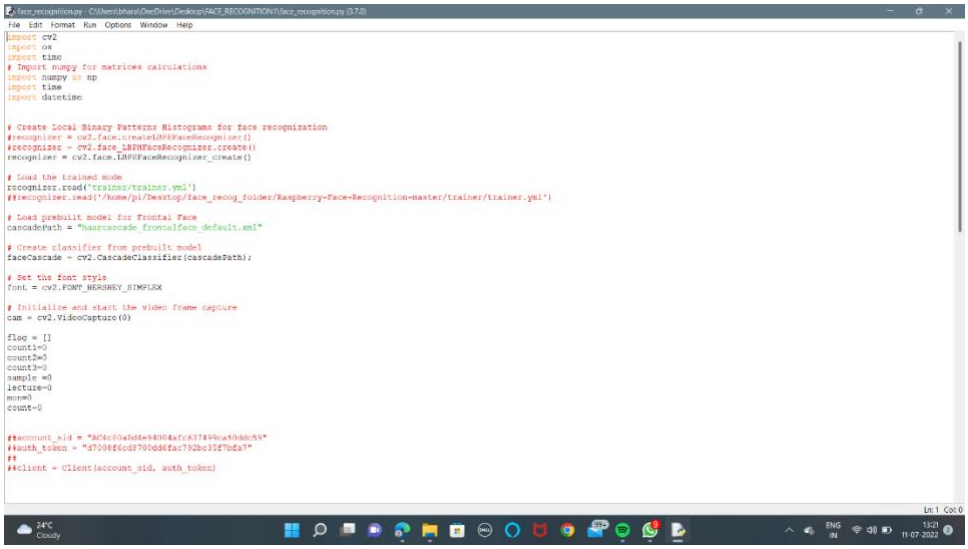
        # Get the image and convert it to grayscale
        PIL_img = Image.open(imagePath).convert('L')

        # PIL image to numpy array
        img_numpy = np.array(PIL_img,'uint8')

        # Get the image id
        id = int(os.path.split(imagePath)[-1].split(".")[1])
        print(id)

        # Get the face from the training images
```

5.3 Snippet code for face recognition



```
import cv2
import os
import time
# Import numpy for matrices calculations
import numpy as np
import time
import datetime


# Create Local Binary Patterns Histograms for face recognization
#recognizer = cv2.face.createLBPHFaceRecognizer()
#recognizer = cv2.face_LBPHFaceRecognizer.create()
recognizer = cv2.face.LBPHFaceRecognizer_create()

# Load the trained mode
recognizer.read('trainer/trainer.yml')
#recognizer.read('/home/pi/Desktop/face_recog_folder/Raspberry-Face-Recognition-master/trainer/trainer.yml')

# Load prebuilt model for Frontal Face
cascadePath = "haarcascade_frontalface_default.xml"

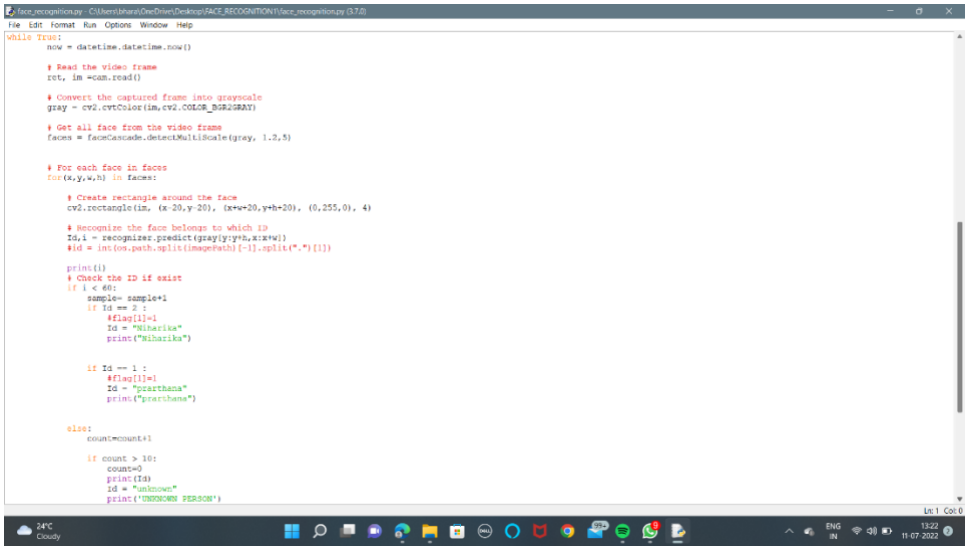
# Create classifier from prebuilt model
faceCascade = cv2.CascadeClassifier(cascadePath);

# Set the font style
font = cv2.FONT_HERSHEY_SIMPLEX

# Initialize and start the video frame capture
cam = cv2.VideoCapture(0)

flag = []
count1=0
count2=0
count3=0
sample =0
lecture=0
mon=0
count=0


##account_sid = "AC6c80a0d4e94084afc637499ca95ddc58"
##auth_token = "d7008f6cd8700dd6fac792bc35f7bfa7"
##
##client = Client(account_sid, auth_token)
```



```
while True:
        now = datetime.datetime.now()

        # Read the video frame
        ret, im =cam.read()

        # Convert the captured frame into grayscale
        gray = cv2.cvtColor(im,cv2.COLOR_BGR2GRAY)

        # Get all face from the video frame
        faces = faceCascade.detectMultiScale(gray, 1.2,5)

        # For each face in faces
        for(x,y,w,h) in faces:

            # Create rectangle around the face
            cv2.rectangle(im, (x-20,y-20), (x+w+20,y+h+20), (0,255,0), 4)

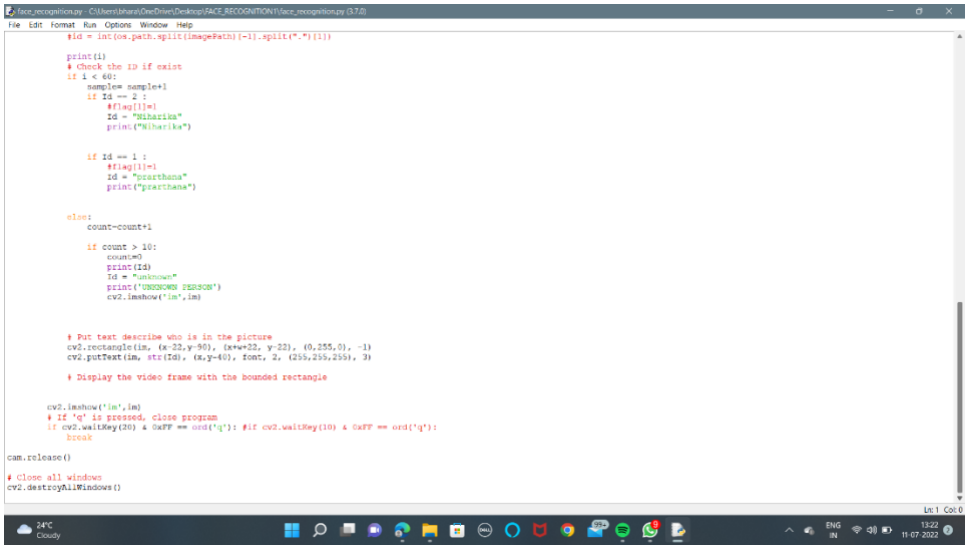
            # Recognize the face belongs to which ID
            Id,i = recognizer.predict(gray[y:y+h,x:x+w])
            #Id = int(os.path.split(imagePath)[-1].split(".")[1])

            print(i)
            # Check the ID if exist
            if i < 60:
                sample= sample+1
                if Id == 2 :
                    #flag[1]=1
                    Id = "Niharika"
                    print("Niharika")


                if Id == 1 :
                    #flag[1]=1
                    Id = "prarthana"
                    print("prarthana")


            else:
                count=count+1

                if count > 10:
                    count=0
                    print(Id)
                    Id = "unknown"
                    print('UNKNOWN PERSON')
```



```
            #Id = int(os.path.split(imagePath)[-1].split(".")[1])

            print(i)
            # Check the ID if exist
            if i < 60:
                sample= sample+1
                if Id == 2 :
                    #flag[1]=1
                    Id = "Niharika"
                    print("Niharika")


                if Id == 1 :
                    #flag[1]=1
                    Id = "prarthana"
                    print("prarthana")


            else:
                count=count+1

                if count > 10:
                    count=0
                    print(Id)
                    Id = "unknown"
                    print('UNKNOWN PERSON')
                    cv2.imshow('im',im)


            # Put text describe who is in the picture
            cv2.rectangle(im, (x-22,y-90), (x+w+22, y-22), (0,255,0), -1)
            cv2.putText(im, str(Id), (x,y-40), font, 2, (255,255,255), 3)

            # Display the video frame with the bounded rectangle


        cv2.imshow('im',im)
        # If 'q' is pressed, close program
        if cv2.waitKey(20) & 0xFF == ord('q'): #if cv2.waitKey(10) & 0xFF == ord('q'):
            break

cam.release()

# Close all windows
cv2.destroyAllWindows()
```

Chapter 6

TESTING

Software testing is an investigation conducted to provide stakeholders with information about the quality of the software product or service under test. Testing is the process of evaluating a system or its component(s) with the intent to find whether it satisfies the specified requirements or not. Testing is executing a system in order to identify any gaps, errors, or missing requirements in contrary to the actual requirements. Software testing can also provide an objective, independent view of the software to allow the business to appreciate and understand the risks of software implementation. Test techniques include the process of executing a program or application with the intent of finding failures. The job of testing is an iterative process as when one fault is fixed, it can illuminate other failures due to deeper faults, or can even create new ones. Software testing can provide objective, independent information about the quality of software and risk of its failure to users or sponsors. Software testing can be conducted as soon as executable software (even if partially complete) exists. The overall approach to software development often determines when and how testing is conducted.

Testing Principle:

Before applying methods to design effective test cases, a software engineer must understand the basic principle that guides software testing. All the tests should be traceable to customer requirements.

6.1 Testing Methods

- **Black-box Testing**

  It is a software testing method in which the internal structure/ design/ implementation of the item being tested is not known to the tester. The examiner doesn't know the structure of the framework, and does not have access to the source code. Usually, while performing this, the tester can interact with the device's user interface, presenting the input and checking the output without knowing where and how it is installed.

- **White-box Testing**

  It is a software testing method in which the internal structure/ design/ implementation of the item being tested is known to the tester. The tester needs to learn the internal functionality of the code to perform a white box test of the application. To check which unit/part of code behaves incorrectly, the tester will look inside the source code.

Levels of Testing:

There are different levels during the process of testing. Levels of testing include different methodologies that can be used while conducting software testing. The main levels of software testing are:

Functional Testing:

This is a type of black-box testing that is based on the specifications of the software that is to be tested. The application is tested by providing input and then the results are examined that need to conform to the functionality it was intended for. Functional testing of software is conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. There are five steps that are involved while testing an application for functionality.

- The determination of the functionality that the intended application is meant to perform.
- The creation of test data based on the specifications of the application.
- The output based on the test data and the specifications of the application.
- The writing of test scenarios and the execution of test cases.
- The comparison of actual and expected results based on the executed test cases.

Non-functional Testing

This section is based upon testing an application from its non-functional attributes. Non-functional testing involves testing software from the requirements which are non-functional in nature but important such as performance, security, user interface, etc. Testing can be done in different levels of SDLC.

Manual and Automation Testing

Manual Testing

Manual testing is the process of manually testing software for defects. It requires a tester to play the role of an end user whereby they use most of the application's features to ensure correct behavior. It is evaluated based on the software requirements. To perform Manual Testing, following steps are performed

1. Read and understand the software project documentation/guides. Also, study the Application Under Test (AUT) if available.
2. Draft Test cases that cover all the requirements mentioned in the documentation.
3. Review and baseline the test cases with Team Lead, Client (as applicable)
4. Execute the test cases on the AUT
5. Report bugs.
6. Once bugs are fixed, again execute the failing test cases to verify they pass.

Automation Testing

Automation Testing or Test Automation is a software testing technique that performs using special automated testing software tools to execute a test case suite. The automation testing software can also enter test data into the System Under Test, compare expected and actual results and generate detailed test reports. Software Test Automation demands considerable investments of money and resources. The goal of Automation is to reduce the number of test cases to be run manually and not to eliminate Manual Testing altogether.

Following steps are followed in an Automation Process

1. Test Tool Selection
2. Define scope of Automation
3. Planning, Design and Development
4. Test Execution
5. Maintenance

6.1.1 Unit Testing

Unit Testing is a type of software testing where individual units or components of a software are tested. The purpose is to validate that each unit of the software code performs as expected. Unit Testing is done during the development (coding phase) of an application by the developers. Unit Tests isolate a section of code and verify its correctness. A unit may be an individual function, method, procedure, module, or object.

Best Practices of Unit Testing:

- Unit Test cases should be independent. In case of any enhancements or change in requirements, unit test cases should not be affected.
- Test only one code at a time.
- Follow clear and consistent naming conventions for your unit tests
- In case of a change in code in any module, ensure there is a corresponding unit Test Case for the module, and the module passes the tests before changing the implementation.
- Adopt a "test as your code" approach. The more code you write without testing, the more paths you have to check for errors.

Table 6.1 UTC Camera Test

TEST CASE	**UTC-1**
TEST NAME	CAMERA TEST
ITEMS BEING TESTED	Capture Image
SAMPLE INPUT	Input Image
EXPECTED OUTPUT	Should Capture the image
ACTUAL OUTPUT	Image capture successful
REMARKS	PASS

Table 6.2 UTC Face Detection

TEST CASE	**UTC-2**
TEST NAME	FACE DETECTION
ITEMS BEING TESTED	Detection of Face
SAMPLE INPUT	Tested for Different Images
Expected Output	Created blob for human face
ACTUAL OUTPUT	FACE DETECTION SUCCESSFUL
REMARKS	PASS

Table 6.3 UTC for Face Recognition

TEST CASE	UTC-3
TEST NAME	Face Recognition
ITEMS BEING TESTED	Face Matching with Database
SAMPLE INPUT	Different Images being tested
Expected Output	Should recognize only stored faces
ACTUAL OUTPUT	Same as expected
REMARKS	PASS

6.1.2 Integration Testing

Integration Testing is defined as a type of testing where software modules are integrated logically and tested as a group. A typical software project consists of multiple software modules, coded by different programmers. The purpose of this level of testing is to expose defects in the interaction between these software modules when they are integrated.

1. **Bottom-up Integration Testing**

   It is a strategy in which the lower-level modules are tested first. These tested modules are then further used to facilitate the testing of higher-level modules. The process continues until all modules at top level are tested. Once the lower-level modules are tested and integrated, then the next level of modules are formed.

2. **Top-down Integration Testing**

   It is a method in which integration testing takes place from top to bottom following the control flow of software system. The higher-level modules are tested first and then

lower-level modules are tested and integrated in order to check the software functionality. Stubs are used for testing if some modules are not ready.

Table 6.4 Integration Test Case 1

TEST CASE	**ITC 1**
TEST NAME	Capture Image and detection of face
ITEMS BEING TESTED	Image capture and face detection
SAMPLE INPUT	Click and select Image
EXCPECTED OUTPUT	Should create blob for face
ACTUAL OUTPUT	Image captured and face detected
REMARKS	PASS

Table 6.5 Integration Test Case 2

TEST CASE	**ITC 2**
TEST NAME	Face recognition and counting
ITEMS BEING TESTED	Images with different face numbers
SAMPLE INPUT	Image input
EXCPECTED OUTPUT	Face recognition and counting
ACTUAL OUTPUT	faces recognized and counted
REMARKS	PASS

6.1.3System testing:

System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black-box testing, and as such, should require no knowledge of the inner design of the code or logic. System testing is important because of the following reasons:

- System testing is the first step in the Software Development Life Cycle, where the application is tested as a whole.
- The application is tested thoroughly to verify that it meets the functional and technical specifications.
- The application is tested in an environment that is very close to the production environment where the application will be deployed.

- System testing enables us to test, verify, and validate both the business requirements as well as the application architecture.

System Testing is shown in below tables

Table 6.6: System Testing Case 1

Sl # Test Case: -	STC-1
Name of Test: -	System testing
Item being tested: -	Attendance Update
Sample Input: -	Input image
Expected output: -	Should detect, count and update attendance
Actual output: -	Attendance Updated Successfully
Remarks: -	Pass

Table 6.7: System Testing Case 2

Sl # Test Case: -	STC-2
Name of Test: -	System testing
Item being tested: -	Alert for absentees
Sample Input: -	Input image
Expected output: -	Should detect, count and update attendance
Actual output: -	Intimated to parent Number Successfully
Remarks: -	Pass

6.1.4 Acceptance Testing

Acceptance testing is actually a series of different tests whose primary purpose is to fully exercise the computer-based system. Include recovery testing crashes, security testing for unauthorized user, etc.

Acceptance testing is sometimes performed with realistic data of the client to demonstrate that the software is working satisfactorily. This testing in FDAC focuses on the external behavior of the system.

Table 6.8: Acceptance Testing Case 1

Test Case ID	Acceptance Test Case 1
Description	Application Should Work in All Platforms
Input	Run Applications Windows XP, Windows 10
Expected output	Functionality should be according to given criteria
Actual Result/Remarks	Working as expected output.
Passed (?)	Yes

6.1.4 Validation Testing

At the culmination of integration testing, software is completely assembled as a package; interfacing errors have been covered and corrected, and final series of software tests-validating testing may begin. Validation can be defined in many ways, but a simple definition is that validation succeeds when software functions in a manner that can be reasonably expected by customers. Reasonable expectation is defined in the software requirement specification- a document that describes all users' visible attributes of the software. The specification contains a section title "validation criteria". Information contained in that section forms the basis for validation testing approach.

Chapter 7

APPLICATIONS AND ADVANTAGES

7.1 Advantages

- Computer human interaction
- Face ID smart cards
- Safety and secure

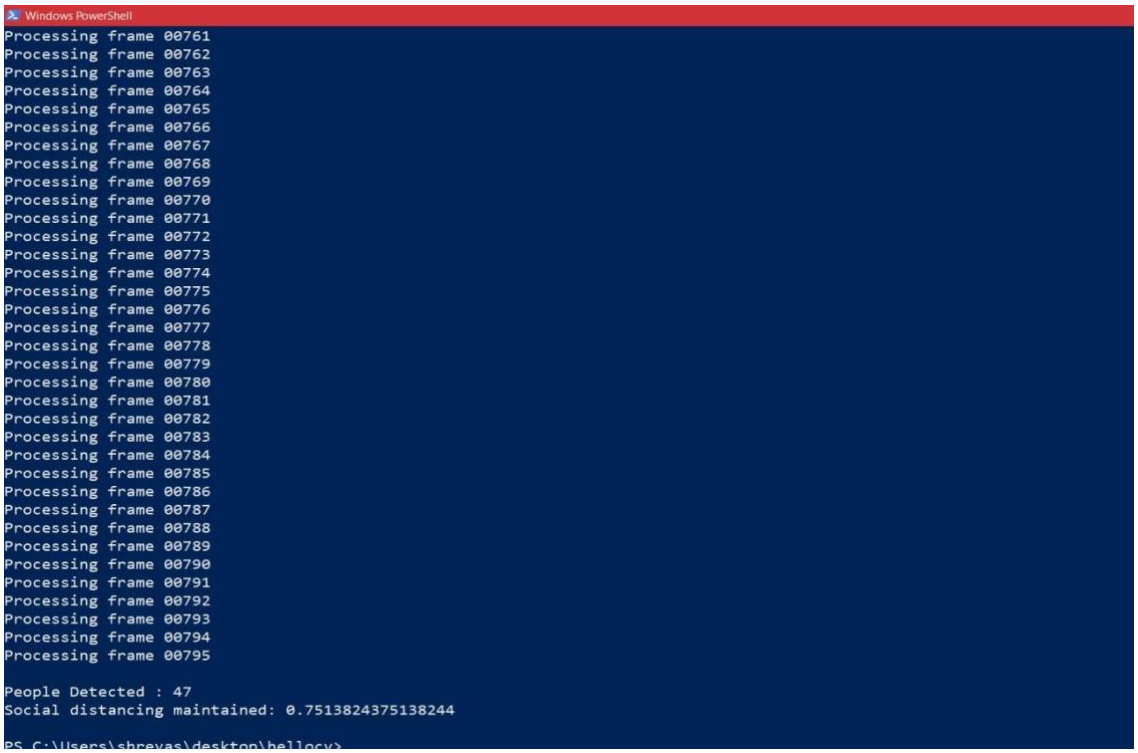
- Audio-visual observation
- Others applications

7.2 Applications

- It saves their time and efforts.
- The software stores the faces that are detected and automatically marks attendance.
- The system is convenient and secure for the user.
- The software can be used for security purposes in organization and secured zones.

Chapter 8

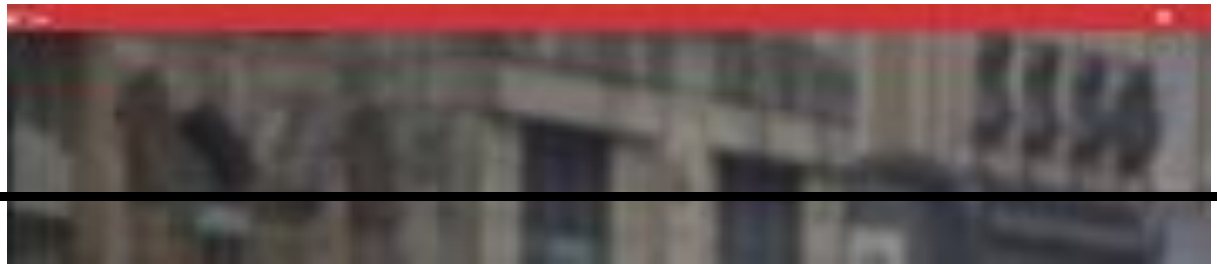
RESULTS



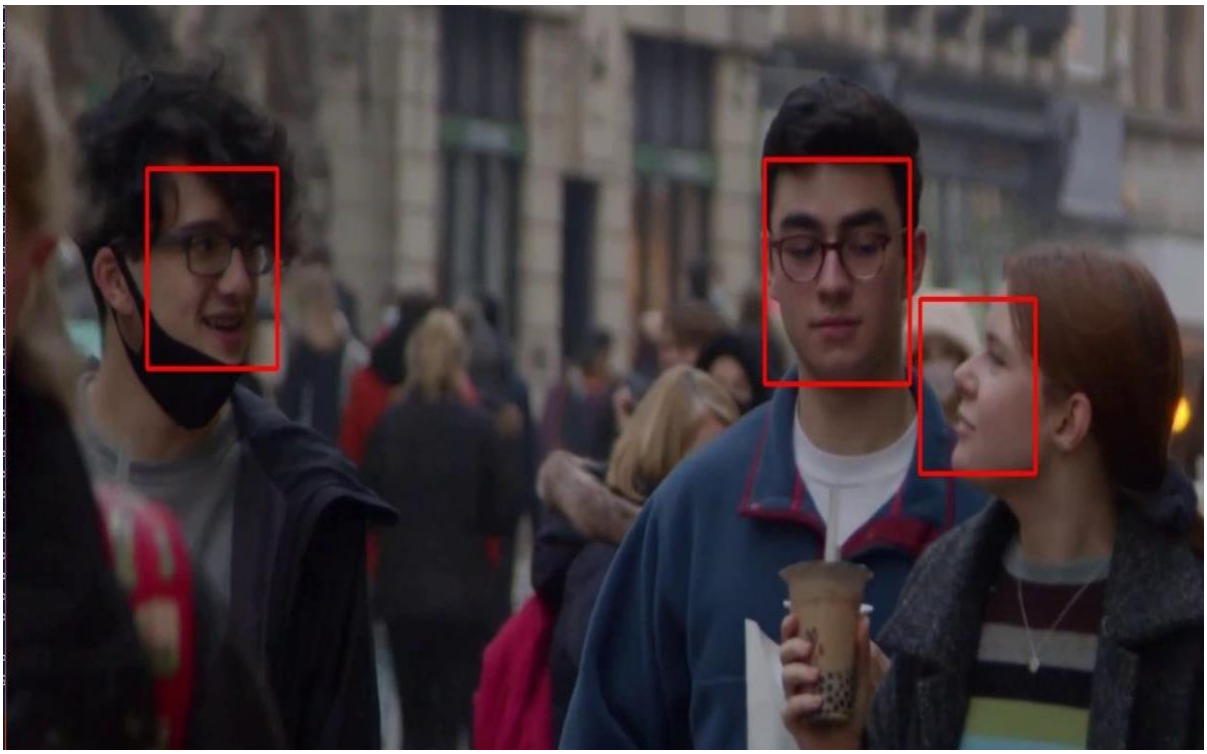
Result Fig 8.1 Taking Datasets without mask



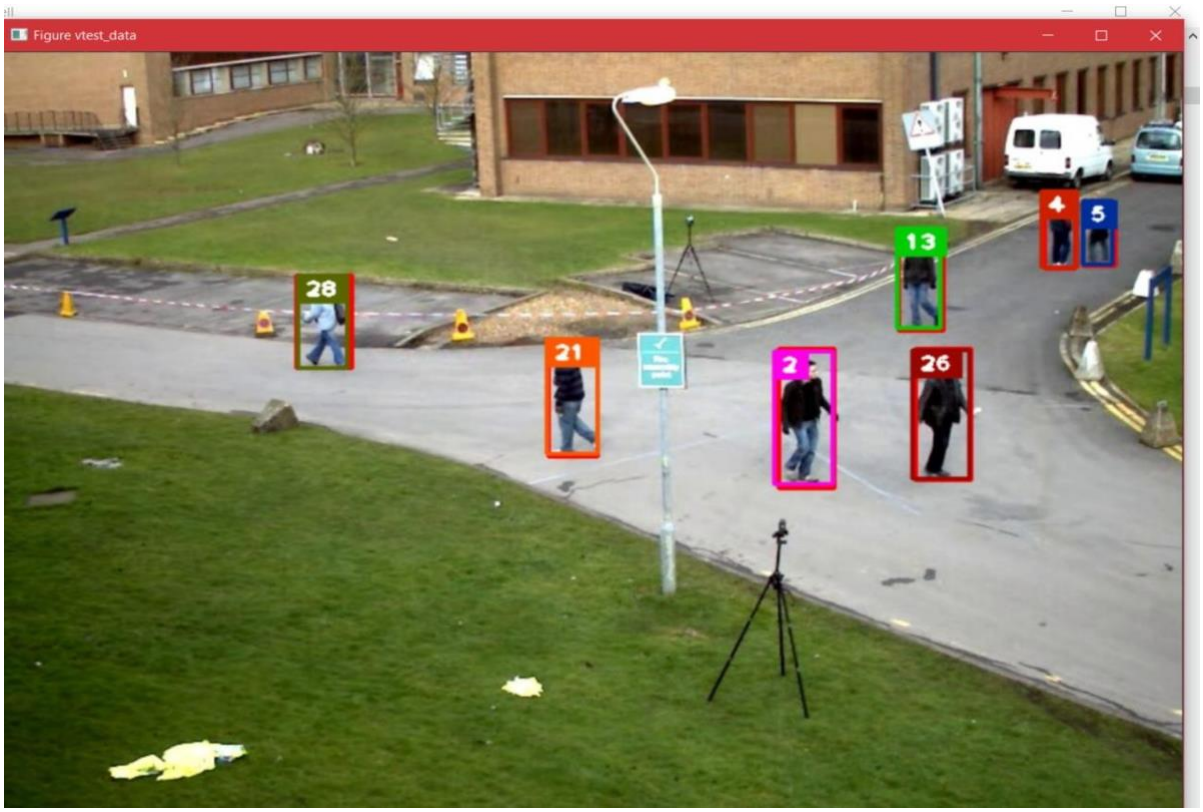
Result Fig 8.2 Taking Datasets with mask



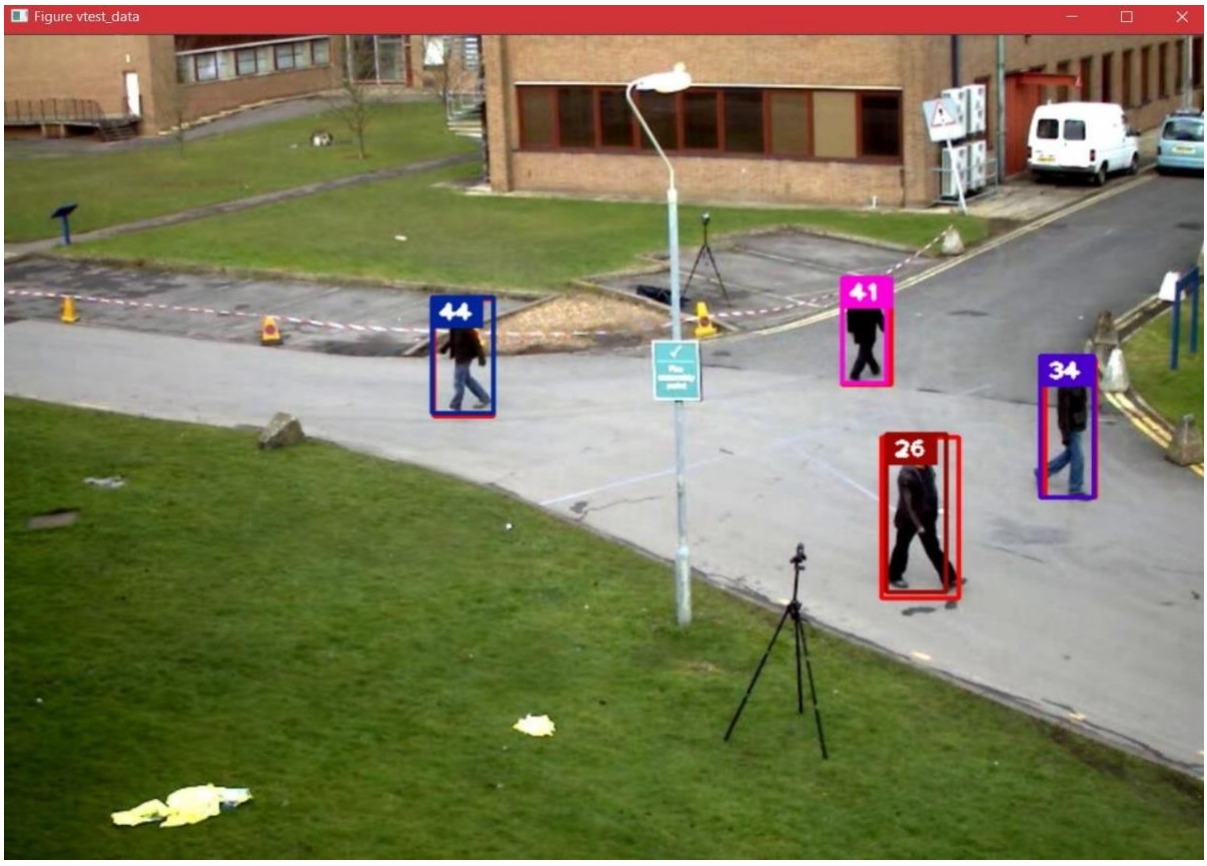
Result Fig 8.3 Training of collected datasets



Result fig 8.4 Recognition of faces



Result Fig 8.5 Recognition of faces



Result Fig 8.6 Recognition of faces

Chapter 9

CONCLUSION AND FUTURE ENHANCEMENT

9.1 Conclusion

Automated Attendance System has been envisioned for the purpose of reducing the errors that occur in the traditional (manual) attendance taking system. The aim is to automate and make a system that is useful to the organization such as an institute. The efficient and accurate method of attendance in the office environment that can replace the old manual methods. This method is secure enough, reliable and available for use. No need for specialized hardware for installing the system in the office. It can be constructed using a camera and computer.

It can be concluded that a reliable, secure, fast and an efficient class attendance management system has been developed replacing a manual and unreliable system. This face detection and recognition system will save time, reduce the amount of work done by the administration and replace the stationery material currently in use with already existent electronic equipment. There is no need for specialized hardware for installing the system as it only uses a computer and a camera. The camera plays a crucial role in the working of the system hence the image quality and performance of the camera in real time scenario must be tested especially if the system is operated from a live camera feed. The system can also be used in permission-based systems and secure access authentication (restricted facilities) for access management, home video surveillance systems for personal security or law enforcement. The major threat to the system is Spoofing. For future enhancements, anti-spoofing techniques like eye blink detection could be utilized to differentiate live from static images in the case where face detection is made from captured images from the classroom. From the overall efficiency of the system i.e., 83.1% human intervention could be called upon to make the system foolproof. A module could thus be included which lists all the unidentified faces and the lecturer is able to manually correct them. Future work could also include adding several well-structured attendance registers for each class and the capability to generate monthly attendance reports and automatically email them to the appropriate staff for review.

Face recognition systems have been grabbing high attention from commercial market point of view as well as pattern recognition field. In this research, Principal component analysis approach to the face recognition problem was studied and a face recognition system based on the eigenfaces approach was proposed. The algorithm has been tested for the standard image databases such as ORL and Indian Face Databases and implemented using Open CV.

The algorithm developed is a generalized one which works well with any type of images and with any size. The tests conducted on PGM and JPEG images of various subjects in different

environments showed that this method gave very good classification of faces though it has limitations over the variations in light, size and in the head orientation. Although the results produced were acceptable, to function as real system, it needs to be more robust and to have other discriminant features. The eigenface approach thus provides a practical solution that is well fitted to the problem of face recognition. It is fast, relatively simple and has been shown to work well in constrained environment.

9.2 Future Enhancements

In future, this system can be used in mobile based face recognition. It can be implemented in real time applications using CCTV camera. This system can be used in hospitals, especially helpful during any pandemic situation. Also, can be installed in business forums, trade centres. Instead of Haar Cascade algorithm, various recognition algorithms can be implemented for effective results. The same project can be utilized for several security applications where authentication is needed to access the privileges of the respective system. It can be used in recognizing guilty parties involving in unauthorized business. Face recognition algorithm can be improved with respect to the utilization of resources so that the project can recognize a greater number of faces at a time which can make the system far better. Many variants of the project can be developed and utilized for home security and personal or organizational benefits. We can also trace a particular student in an organization quickly with the help of this system. The speed of RAM can improve. More features can be extracted or hybrid feature method can be used for face recognition. Database management languages can be used for online availability of attendance record which can be accessed by parents and students individually. Mobile application software can be developed for easy assessment of records. Multi- cameras can be used so that efficiency can be increased.

Chapter 10

REFERENCE

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